

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\403-E-002ABCHydrogen Compressor 2nd stage interstage Cooler.pvdb

Date of Analysis : Aug 12,2026 9:55am

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 2 of 170
 Vessel Design Summary: Step: 30 9:55am Aug 12,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 828.000 mm. ID
 Vessel Design Length, Tangent to Tangent 7383.50 mm.
 Specified Datum Line Distance 50.00 mm.
 Shell Side Design Temperature 143 °C
 Channel Side Design Temperature 71 °C
 Shell Side Design Pressure 12.653 bars
 Channel Side Design Pressure 5.516 bars
 Shop Shell Side Test Pressure 16.948 bars
 Field Shell Side Test Pressure 16.449 bars
 Shop Channel Side Test Pressure 17.003 bars
 Field Channel Side Test Pressure 7.171 bars

Wind Design Code IS 875 (PART 3) Amd. 1&2 (2020)
 Earthquake Design Code IS 1893 RSM Amd. 1&2 (2018)

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-266 4	...	...	K03017	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 4	...	...	K03017	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 4	...	...	K03017	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Chnl. head	5.595	0.00	29.000	3.2000	Yes	No
Chnl. rear	5.595	0.00	29.000	3.2000	N/A	No
Chnl. rear flange-2	5.516	0.00	17.000	3.2000	N/A	No
Shell flange	12.653	0.00	18.000	0.0000	N/A	No
Shell	12.653	0.00	35.000	3.2000	N/A	No
shell head	12.653	0.00	23.000	3.2000	No	No

Liquid Level: 800.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.005

Element Types and Properties:

Element	"To" Elev	Element	Nominal	Finished	Reqd Thk	Reqd Thk	Long	Circ
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 FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 3 of 170
 Vessel Design Summary: Step: 30 9:55am Aug 12,2026

Type	mm.	Length mm.	Thickness mm.	Thickness mm.	Internal mm.	External mm.	Eff	Eff
Ellipse	0.0	50.0	12.0	12.0	5.7	...	1.00	1.00
Cylinder	700.0	700.0	12.0	12.0	5.7	...	1.00	1.00
Body Flg	799.5	95.0	85.0	77.0	72.1	72.1	1.00	1.00
Body Flg	983.5	84.0	68.0	60.0	53.5	53.5	1.00	1.00
Cylinder	7283.5	6300.0	14.0	14.0	7.0	...	1.00	1.00
Ellipse	7333.5	50.0	12.0	10.2	7.0	...	1.00	1.00

Local Stress Analysis Results:

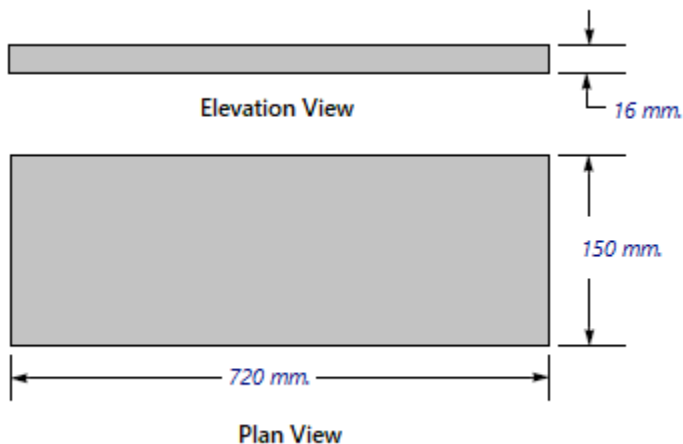
Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N2- Channel ou	WRC-297	0.925	Shell	Passed
N1- Channel in	WRC-297	0.928	Shell	Passed
Shell inlet	WRC-297	0.928	Shell	Passed
Shell outlet	WRC-297	0.967	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	175.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	660.000 mm.
Wear Pad Width	225.000 mm.
Wear Pad Thickness	12.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	1260.000 mm.
Baseplate Length	720.000 mm.
Baseplate Thickness	16.000 mm.
Baseplate Width	150.000 mm.
Number of Ribs (including outside ribs)	3
Rib Thickness	16.000 mm.
Web Thickness	16.000 mm.
Height of Center Web	306.000 mm.
Number of Bolts in Baseplate	2

Baseplate Sketch





PV Elite 28 Licensee: SL User
FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 4 of 170
Vessel Design Summary: Step: 30 9:55am Aug 12,2026

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	8896.26	Kgf
Maximum Transverse Saddle Shear Load	792.30	Kgf
Maximum Longitudinal Saddle Shear Load	820.84	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	8330.74	Kgf
Maximum Transverse Saddle Shear Load	261.46	Kgf
Maximum Longitudinal Saddle Shear Load	60.87	Kgf

Weights:

Fabricated - Bare w/o removable internals	7601.4	kg.
Shop Test - Fabricated + water (full)	11261.7	kg.
Shipping - Fab. + rem. intls.+ shipping app.	7601.4	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	7601.4	kg.
Empty - Fab. + intls. + details + wghts.	7657.3	kg.
Operating - Empty + operating liquid (no ca)	8950.7	kg.
Field Test - Empty weight + test medium (full)	10821.3	kg.

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PV Elite 28 Licensee: SL User
 FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 5 of 170
 Nozzle Summary: Step: 28 9:55am Aug 12,2026

Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N2- Channel out	29	...	...	OK	5.70	OK	Passed
N1- Channel inl	29	...	...	OK	5.70	OK	Passed
Shell inlet	13	...	...	OK	7.05	OK	Passed
Shell outlet	32	...	...	OK	7.05	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 13 Nozzle : Shell inlet [Shellside]
 Minimum MAWP Nozzles : 29 Nozzle : N2- Channel out [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
20	N2- Channel outl	350.000	270.000	400.150
20	N1- Channel inle	350.000	90.000	400.150
50	Shell inlet	1533.500	270.000	418.235
50	Shell outlet	6933.500	90.000	498.550

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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 Nozzle Schedule: Step: 27 9:55am Aug 12,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N2- Channel outl	8.000 in	80	WNF	8.625	12.700	319.08	12.00	276.75	150
N1- Channel inle	8.000 in	80	WNF	8.625	12.700	319.08	12.00	276.75	150
Shell inlet	8.000 in	40	WNF	8.625	8.179	...	...	278.75	300
Shell outlet	10.000 in	80	WNF	10.750	15.088	...	...	287.16	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N2- Channel	SA-106 B	12.000	30.000	10.000	12.000	...
N1- Channel	SA-106 B	12.000	30.000	10.000	12.000	...
Shell inlet	SA-266 4	6.000	10.000	...	...	...
Shell outle	SA-266 4	6.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
N2- Channel outl	300.000	270.0	250.00	0.00	Chnl. rear
N1- Channel inle	300.000	90.0	250.00	0.00	Chnl. rear
Shell inlet	1483.500	270.0	250.00	0.00	Shell
Shell outlet	6883.500	90.0	250.00	0.00	Shell

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PV Elite 28 Licensee: SL User
FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 7 of 170
Bill of Material: Step: 26 9:55am Aug 12,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 12.0mm. THK X 828.0mm. ID X 50.0mm.	SA-516 70
1	WELD NECK FLANGE: 85.0mm. THK X 1020.0mm. OD	SA-266 4
1	WELD NECK FLANGE: 68.0mm. THK X 1020.0mm. OD	SA-266 4
6	BAFFLES: 780.0mm DIA X 6.0mm THK	SA-36
1	TUBESHEET: 1020.0mm X 100.0mm THK	SA-266 4
370	TUBES: 6096.0mm X 25.4mm DIA X 2.5mm THK	SA-179
2	SADDLE: 175.0mm X 120 DEG	SA-516 70
1	INSULATION: 6300mm X 25mm THK	Hot
1	INSULATION: 224mm X 25mm THK	Hot
1	GASKET: 886.0mm. OD X 860.0mm. ID	...
24	BODY FLANGE BOLTS: 25.4mm. DIA	SA-193 B7
48	NUTS FOR BODY FLANGE BOLTS: 25.4mm. DIA	...
2	CYLINDRICAL SEGMENT 12.0mm. THK X 1033.3mm. X 700.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 12.0mm. THK X 534.7mm. X 700.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm. THK X 2088.2mm. X 6300.0mm.	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm. THK X 513.0mm. X 6300.0mm.	SA-516 70
1	NAMEPLATE	...



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 FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 8 of 170
 MDMT Summary: Step: 29 9:55am Aug 12,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Notes								
Shell flange [11]	B	-14	-48	-29	0.371	16.000	1.00	No
Shell [8]	B	-18	-48	-29	0.367	14.000	1.00	No
shell head [10]	B	-28	-48	-29	0.558	10.200	1.00	No
shell head [7]	B	-23	-48	-29	0.450	12.000	1.00	No
Left Saddle [24]	B	-23	-48	-29	0.367	12.000	1.00	No
Right saddle [24]	B	-23	-48	-29	0.367	12.000	1.00	No
Shell inlet [1]	B	-18	-48	-29	0.356	14.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
Shell outlet [1]	B	-18	-48	-29	0.356	14.000	1.00	No
Nozzle Flg [5]	A	-18	-104					
Tubesheet: SS[13]	B	-1	-25	-29	0.576	25.000	1.00	No
Warmest MDMT:		-1	-25					
Chnl. rear fl[11]	B	-23	-48	-29	0.455	12.000	1.00	No
Chnl. head [10]	B	-23	-48	-29	0.445	12.000	1.00	No
Chnl. head [7]	B	-23	-48	-29	0.452	12.000	1.00	No
Chnl. rear [8]	B	-23	-48	-29	0.452	12.000	1.00	No
N2- Channel ou[1]	B	-5	-104	-29	0.065	11.113	1.00	No
Nozzle Flg [5]	A	-18	-104					
N1- Channel in[1]	B	-5	-104	-29	0.066	11.113	1.00	No
Nozzle Flg [5]	A	-18	-104					
Tubesheet: CS[14]	B	-1	-25	-29	0.576	25.000	1.00	No
Bolting [21]		-48						
Warmest MDMT:		-1	-25					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	11.4	Pass
Channel/Tube	-29.0	11.4	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.



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FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 9 of 170
MDMT Summary: Step: 29 9:55am Aug 12,2026

UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:
Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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PV Elite 28 Licensee: SL User
FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 10 of 170
Warnings and Errors: Step: 0 9:55am Aug 12,2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: Chnl. rear flange and Shell flange

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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PV Elite 28 Licensee: SL User
 FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 11 of 170
 Input Echo: Step: 1 9:55am Aug 12,2026

Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	12.653	bars
Channel Side Design Pressure	5.516	bars
Shell Side Design Temperature	143.3	°C
Channel Side Design Temperature	71.1	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Severe Sour	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	11.4	°C
MDMT (CET), Tube Side	11.4	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070



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 Input Echo: Step: 1 9:55am Aug 12,2026

0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vortex Shedding N

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	IS 875 (PART 3):2015-2020	
Basic Wind Speed for IS-875	180	Km/hr
Wind Zone Number	5	
Base Elevation	300	mm.
Percent Wind for Hydrotest	33.0	
Risk Factor	[k1] 1.0	
Terrain Category	2	
Equipment Class	Class A	
Topography Factor	[k3] 1.0	
Importance Factor	[k4] 1.15	
Damping Factor (Beta) for Wind (Ope)	0.0100	



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 FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 13 of 170
 Input Echo: Step: 1 9:55am Aug 12,2026

Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Seismic Design Code	IS 1893-RSM
Importance Factor for IS-1893	2.000
Soil Factor	1.000
Zone Number	2.000
Percent Seismic for Hydrotest	0.000
Period of Vibration (User defined) Sec	0.000
Damping Factor	2.000

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Consider Vertical Acceleration	Yes
Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N
Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:



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Element#	1/6	
Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	Chnl. head	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	828	mm.
Element Thickness	12	mm.
Internal Corrosion Allowance	3.2	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	5.516	bars
Design Temperature Internal Pressure	71	°C
Design External Pressure	0	bars
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	244.78	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Cooling water	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	800	mm.
Liquid Density	0.0009984	kg./cm ³

Element#	2/6	
Element From Node	20	
Element To Node	30	
Element Type	Cylinder	
Description	Chnl. rear	
Distance "FROM" to "TO"	700	mm.
Inside Diameter	828	mm.
Element Thickness	12	mm.
Internal Corrosion Allowance	3.2	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	5.516	bars
Design Temperature Internal Pressure	71	°C
Design External Pressure	0	bars
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	20	
Detail Type	Liquid	



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Detail ID	Cooling water	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	800	mm.
Liquid Density	0.0009984	kg./cm ³

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N2- Channel outl	
Dist. from "FROM" Node / Offset dist	300	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	39.853	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	N1- Channel inle	
Dist. from "FROM" Node / Offset dist	300	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	39.853	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Weight	
Detail ID	Partition plate	
Dist. from "FROM" Node / Offset dist	0	mm.
Miscellaneous Weight	70	Kgf
Offset from Element Centerline	0	mm.

Element#	3/6	
Element From Node	30	
Element To Node	40	
Element Type	Flange	
Description	Chnl. rear flange-2	
Distance "FROM" to "TO"	95	mm.
Flange Inside Diameter	828	mm.
Element Thickness	77	mm.
Internal Corrosion Allowance	3.2	mm.
Nominal Thickness	85	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	5.516	bars
Design Temperature Internal Pressure	71	°C
Design External Pressure	0	bars
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 4	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	225	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm.
Yield Stress, Operating	231.95	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	



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UNS Number	K03017
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	4/6
Element From Node	40
Element To Node	50
Element Type	Flange
Description	Shell flange
Distance "FROM" to "TO"	84 mm.
Flange Inside Diameter	828 mm.
Element Thickness	60 mm.
Internal Corrosion Allowance	0 mm.
Nominal Thickness	68 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	12.653 bars
Design Temperature Internal Pressure	143 °C
Design External Pressure	0 bars
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 4
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element#	5/6
Element From Node	50
Element To Node	60
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	6300 mm.
Inside Diameter	828 mm.
Element Thickness	14 mm.
Internal Corrosion Allowance	3.2 mm.
Nominal Thickness	14 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	12.653 bars
Design Temperature Internal Pressure	143 °C
Design External Pressure	0 bars
Design Temperature External Pressure	40 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	232.8 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	50
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	1260 mm.
Width of Saddle	175 mm.



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Height of Saddle at Bottom	660	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	225	mm.
Thickness of Wear Plate	12	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	50	
Detail Type	Saddle	
Detail ID	Right saddle	
Dist. from "FROM" Node / Offset dist	5040	mm.
Width of Saddle	175	mm.
Height of Saddle at Bottom	660	mm.
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm.
Width of Wear Plate	225	mm.
Thickness of Wear Plate	12	mm.
Contact Angle, Wear Plate (degrees)	132.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm.
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm.

Element From Node	50	
Detail Type	Liquid	
Detail ID	Hydrogen	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	800	mm.
Liquid Density	0.49943E-05	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	6300	mm.
Thickness of Insulation	25	mm.
Density	0.00012	kg./cm ³

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Shell inlet	
Dist. from "FROM" Node / Offset dist	500	mm.
Nozzle Diameter	8	in.
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	51.383	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Shell outlet	
Dist. from "FROM" Node / Offset dist	5900	mm.
Nozzle Diameter	10	in.
Nozzle Schedule	80	



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Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	89.766	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 4	

Element#	6/6	
Element From Node	60	
Element To Node	70	
Element Type	Elliptical	
Description	shell head	
Distance "FROM" to "TO"	50	mm.
Inside Diameter	828	mm.
Element Thickness	10.2	mm.
Internal Corrosion Allowance	3.2	mm.
Nominal Thickness	12	mm.
External Corrosion Allowance	0	mm.
Design Internal Pressure	12.653	bars
Design Temperature Internal Pressure	143	°C
Design External Pressure	0	bars
Design Temperature External Pressure	40	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	60	
Detail Type	Liquid	
Detail ID	Hydrogen	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Liquid	800	mm.
Liquid Density	0.49943E-05	kg./cm ³
Element From Node	60	
Detail Type	Insulation	
Detail ID	Hot insulation	
Dist. from "FROM" Node / Offset dist	0	mm.
Height/Length of Insulation	225	mm.
Thickness of Insulation	25	mm.
Density	0.00012	kg./cm ³



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XY Coordinate Calculations: Step: 2 9:55am Aug 12,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Chnl. head		50	...	50	...
Chnl. rear		750	...	700	...
Chnl. rear flan		849.5	...	95	...
Shell flange		1033.5	...	84	...
Shell		7333.5	...	6300	...
shell head		7383.5	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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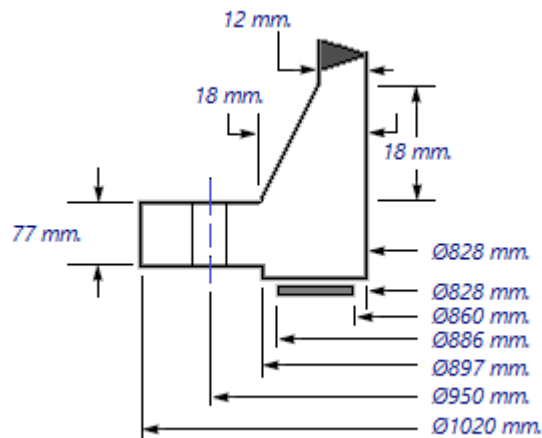
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 Flg Calc [Int P]: New Flange Flng: 11 9:55am Aug 12,2026

Flange Input Data Values

Description: New Flange :

Chnl. rear flange-2

Flange Type	Integral Weld Neck		
Design Pressure	P	5.52	bars
Design Temperature		71	°C
Internal Corrosion Allowance	ci	3.2000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	828.000	mm.
Flange Outside Diameter	A	1020.000	mm.
Flange Thickness	t	77.0000	mm.
Thickness of Hub at Small End	go	12.0000	mm.
Thickness of Hub at Large End	gl	18.0000	mm.
Length of Hub	h	18.0000	mm.
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	950.000	mm.
Nominal Bolt Diameter	a	25.4000	mm.
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	897.000	mm.
Flange Face Inside Diameter	Fid	828.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	886.000	mm.
Gasket Inside Diameter	Gi	860.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.





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 Flg Calc [Int P]: New Flange Flng: 11 9:55am Aug 12,2026

ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (5.52 \cdot (828.0/2 + 3.2)) / (138.0 \cdot 1.0 - 0.6 \cdot 5.52) + Ca$$

$$= 4.8717 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.8) / (417.2 + 0.6 \cdot 8.8)$$

$$= 28.742 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 12.0) / (414.0 + 0.6 \cdot 12.0)$$

$$= 39.313 \text{ bars}$$

Corroded Flange Thickness, $t_c = T - c_i$	73.800 mm.
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	834.400 mm.
Corroded Large Hub, $g1_{cor} = g1 - c_i$	14.800 mm.
Corroded Small Hub, $g0_{cor} = g0 - c_i$	8.800 mm.
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	43.000 mm.

Gasket Contact Width, $N = (G_o - G_i)/2$	13.000 mm.
Basic Gasket Width, $bo = N/2$	6.500 mm.
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	6.425 mm.
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	873.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 873.1509^2 \cdot 5.516$$

$$= 33681.332 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 873.1509 \cdot 3.0 \cdot 5.52$$

$$= 5947.774 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 834.4^2 \cdot 5.516$$

$$= 30758.088 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 33681 - 30758$$

$$= 2923.246 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(33681 + 5948 + 0, 0)$$

$$= 39629.105 \text{ Kg}$$

$$= 90904.117 \text{ Kg}, \text{ Mating Flange Load Governs}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 873.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 123904.766 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/S_b, Wg/S_a)$$

$$= \max((90904 + 0 + 4 \cdot 0/873.15)/172, 123905/172)$$

$$= 70.647 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t/(m + 0.5)$$

$$= 2 \cdot 25.4 + 6 \cdot 73.8/(3.0 + 0.5)$$

$$= 177.314 \text{ mm.}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 950.0 \cdot \sin(\pi/24)$$

$$= 124.000 \text{ mm.}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{(Bs/(2a + t))}, 1)$$

$$= \max(\sqrt{(124.0/(2 \cdot 25.4 + 73.8))}, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	70.647	85.316	
Radial Distance between Hub and Bolts:	34.925	43.000	
Radial Distance between Bolts and Edge:	26.987	35.000	
Circ. Spacing between the Bolts:	57.150	124.000	177.314

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi(Go + Gi))$$

$$= 85.316 \cdot 172.0 / (68.95 \cdot \pi(886.0 + 860.0))$$

$$= 3.880 \text{ mm.}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa(Am + Ab) / 2$$

$$= 172.0(70.6474 + 85.316) / 2$$

$$= 136768.00 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo - H$$

$$= 90904 - 33681$$

$$= 57222.79 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (950.0 - 873.1509) / 2$$

$$= 38.4246 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((950.0 - 834.4) / 2 + 38.4246)$$

$$= 48.1123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (950.0 - 834.4 - 14.8) / 2$$

$$= 50.4000 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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 Flg Calc [Int P]: New Flange Flng: 11 9:55am Aug 12,2026

End Pressure, Md	30758.	50.4000	1.0000	1550.
Face Pressure, Mt	2923.	48.1123	1.0000	141.
Gasket Load, Mg	57223.	38.4246	1.0000	2199.
Gasket Seating, Matm	136768.	38.4246	1.0000	5255.

Total Moment for Operation, Mo 3890. Kg-m.
 Total Moment for Gasket seating, Mg 5255. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 85.690 mm.
 Hub Ratio, h/h0 = HL / H0 0.210
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.682

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.896
 Factor V per 4.16.5 0.384
 Factor f per 4.16.5 1.835

Stress factor from Table 4.16.4 [K]:

= A / B
 = 1020.0/834.4
 = 1.222

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K))) / (K² - 1))/(K - 1)
 = (0.66845 + (5.7169(1.222² • log10(1.222)))/(1.222² - 1))/(1.222 - 1)
 = 9.782

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1)/((1.0472 + 1.9448 • K²) • (K - 1))
 = (1.222²(1 + 8.55246 • log10(1.222)) - 1)/((1.0472 + 1.9448 • 1.222²) • (1.222 - 1))
 = 1.830

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1)/(1.36136(K² - 1) • (K - 1))
 = (1.222²(1 + 8.55246 • log10(1.222)) - 1)/(1.36136(1.222² - 1) • (1.222 - 1))
 = 10.749

Stress factor from Table 4.16.4 [Z]:

= (K² + 1)/(K² - 1)
 = (1.222² + 1)/(1.222² - 1)
 = 5.046

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
 = 10.749 • 8.8² • 85.69/0.384
 = 185629.031 mm.³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.896/85.69
 = 0.010 1/mm.

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
 = (73.8 • 0.01 + 1)/1.83 + 73.8³/185629.031
 = 3.133

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
 = (1.8355•3890/834.4)/(3.1334•14.8²)
 = 122.25 N./mm²



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 Flg Calc [Int P]: New Flange Flng: 11 9:55am Aug 12,2026

Longitudinal Hub Stress, Seating [SHa]:
 $= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g1^2)$
 $= (1.8355 \cdot 5255 / 834.4) / (3.1334 \cdot 14.8^2)$
 $= 165.18 \text{ N./mm}^2$

Radial Flange Stress, Operating [SRo]:
 $= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$
 $= (2.0287 \cdot 3890 / 834.4) / (3.1334 \cdot 73.8^2)$
 $= 5.43 \text{ N./mm}^2$

Radial Flange Stress, Seating [SRa]:
 $= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$
 $= (2.0287 \cdot 5255 / 834.4) / (3.1334 \cdot 73.8^2)$
 $= 7.34 \text{ N./mm}^2$

Tangential Flange Stress, Operating [STo]:
 $= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$
 $= (9.7819 \cdot 3890 / (73.8^2 \cdot 834.4)) - 5.0457 \cdot 5$
 $= 54.69 \text{ N./mm}^2$

Tangential Flange Stress, Seating [STa]:
 $= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$
 $= (9.7819 \cdot 5255 / (73.8^2 \cdot 834.4)) - 5.0457 \cdot 7$
 $= 73.89 \text{ N./mm}^2$

Average Flange Stress, Operating [SAo]:
 $= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$
 $= (122 + \max(5.55)) / 2$
 $= 88.47 \text{ N./mm}^2$

Average Flange Stress, Seating [SAa]:
 $= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$
 $= (165 + \max(7.74)) / 2$
 $= 119.53 \text{ N./mm}^2$

Bolt Stress, Operating [BSo]:
 $= \text{Wm1} / \text{Ab}$
 $= 90904 / 85.316$
 $= 104.49 \text{ N./mm}^2$

Bolt Stress, Seating [BSa]:
 $= (\text{Wm2} / \text{Ab})$
 $= (123905 / 85.316)$
 $= 142.43 \text{ N./mm}^2$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	122.25	207.00	165.18 207.00
Radial Flange	5.43	138.00	7.34 138.00
Tangential Flange	54.69	138.00	73.89 138.00
Maximum Average	88.47	138.00	119.53 138.00
Bolting	104.49	172.00	142.43 172.00

Minimum Required Flange Thickness 72.111 mm.

Estimated M.A.W.P. (Operating) 17 bars
 Estimated Finished Weight of Flange at given Thk. 171.9 kg.
 Estimated Unfinished Weight of Forging at given Thk 205.2 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:



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$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5348.7 / 1.0178 \cdot 9806.637 \cdot 0.384 / (2.702 \cdot 202713 \cdot 8.8^2 \cdot 85.69 \cdot 0.3)$$

$$= 0.946 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 3958.8 / 1.0178 \cdot 9806.637 \cdot 0.384 / (2.702 \cdot 199849 \cdot 8.8^2 \cdot 85.69 \cdot 0.3)$$

$$= 0.710 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5255.4 / 1.0 \cdot 9806.637 \cdot 0.384 / (3.133 \cdot 202713 \cdot 8.8^2 \cdot 85.69 \cdot 0.3)$$

$$= 0.816 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 3889.7 / 1.0 \cdot 9806.637 \cdot 0.384 / (3.133 \cdot 199849 \cdot 8.8^2 \cdot 85.69 \cdot 0.3)$$

$$= 0.613 \text{ (should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.455, Temperature Reduction per Fig. UCS 66.1 = 39 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)-(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	101.54	207.00	135.35	207.00
Radial Flange	6.82	138.00	9.09	138.00
Tangential Flange	41.08	138.00	54.76	138.00
Maximum Average	71.31	138.00	95.05	138.00
Bolting	104.49	172.00	142.43	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 61.900 mm.

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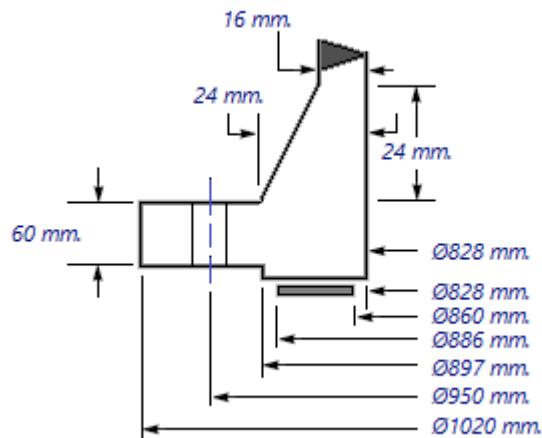
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Flange Input Data Values

Description: New Flange :

Shell flange

Flange Type	Integral Weld Neck		
Design Pressure	P	12.65	bars
Design Temperature		143	°C
Internal Corrosion Allowance	ci	0.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	828.000	mm.
Flange Outside Diameter	A	1020.000	mm.
Flange Thickness	t	60.0000	mm.
Thickness of Hub at Small End	go	16.0000	mm.
Thickness of Hub at Large End	g1	24.0000	mm.
Length of Hub	h	24.0000	mm.
Flange Material		SA-266 4	
Flange Material UNS number		K03017	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	950.000	mm.
Nominal Bolt Diameter	a	25.4000	mm.
Type of Threads	TEMA Thread Series		
Number of Bolts		24	
Flange Face Outside Diameter	Fod	897.000	mm.
Flange Face Inside Diameter	Fid	828.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	886.000	mm.
Gasket Inside Diameter	Gi	860.000	mm.
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	4.5000	mm.
Raised face depth	Fd	0.0000	mm.





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E \cdot 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (12.65 \cdot (828.0/2 + 0.0)) / (138.0 \cdot 1.0 \cdot 0.6 \cdot 12.65) + Ca$$

$$= 3.8172 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (414.0 + 0.6 \cdot 16.0)$$

$$= 52.121 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 16.0) / (414.0 + 0.6 \cdot 16.0)$$

$$= 52.121 \text{ bars}$$

The hub small end thickness is now taken as the attached shell thickness.

Corroded Flange ID,	Bcor = B+2*ci	828.000 mm.
Corroded Large Hub,	g1Cor = g1-ci	24.000 mm.
Corroded Small Hub,	g0Cor = g0-ci	14.000 mm.
Code R Dimension,	R = (C-B)/2 - g1	37.000 mm.
Gasket Contact Width,	N = (Go - Gi)/2	13.000 mm.
Basic Gasket Width,	bo = N/2	6.500 mm.
Effective Gasket Width,	b = Cb*sqrt(bo)	6.425 mm.
Gasket Reaction Diameter,	G = Go - 2 * b	873.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 873.1509^2 \cdot 12.653$$

$$= 77260.680 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 6.4246 \cdot \pi \cdot 873.1509 \cdot 3.0 \cdot 12.65$$

$$= 13643.436 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot Bcor^2 \cdot P$$

$$= \pi/4 \cdot 828.0^2 \cdot 12.653$$

$$= 69476.938 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - Hd$$

$$= 77261 - 69477$$

$$= 7783.740 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(77261 + 13643 + 0, 0)$$

$$= 90904.117 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 6.4246 \cdot \pi \cdot 873.151 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 123904.766 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((90904 + 0 + 4 \cdot 0 / 873.15) / 172, 123905 / 172)$$

$$= 70.647 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 25.4 + 6 \cdot 60.0/(3.0 + 0.5)$
 $= 153.657 \text{ mm.}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 950.0 \cdot \sin(\pi/24)$
 $= 124.000 \text{ mm.}$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 $= \max(\sqrt{Bs/(2a + t)}, 1)$
 $= \max(\sqrt{124.0/(2 \cdot 25.4 + 60.0)}, 1)$
 $= 1.0579$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	70.647	85.316	
Radial Distance between Hub and Bolts:	34.925	37.000	
Radial Distance between Bolts and Edge:	26.987	35.000	
Circ. Spacing between the Bolts:	57.150	124.000	153.657

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 85.316 \cdot 172.0/(68.95 \cdot \pi(886.0 + 860.0))$
 $= 3.880 \text{ mm.}$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
 $= Sa(Am + Ab)/2$
 $= 172.0(70.6474 + 85.316)/2$
 $= 136768.00 \text{ Kg}$

Gasket Load for the Operating Condition [HG]:
 $= Wo - H$
 $= 90904 - 77261$
 $= 13643.44 \text{ Kg}$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G)/2$
 $= (950.0 - 873.1509)/2$
 $= 38.4246 \text{ mm.}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5((C - B)/2 + hg)$
 $= 0.5((950.0 - 828.0)/2 + 38.4246)$
 $= 49.7123 \text{ mm.}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1)/2$
 $= (950.0 - 828.0 - 24.0)/2$
 $= 49.0000 \text{ mm.}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	69477.	49.0000	1.0579	3602.
Face Pressure, Mt	7784.	49.7123	1.0579	409.
Gasket Load, Mg	13643.	38.4246	1.0579	555.
Gasket Seating, Matm	136768.	38.4246	1.0579	5560.

Total Moment for Operation, Mo 4565. Kg-m.
 Total Moment for Gasket seating, Mg 5560. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 107.666 mm.
 Hub Ratio, h/h0 = HL / H0 0.223
 Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.714

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.893
 Factor V per 4.16.5 0.373
 Factor f per 4.16.5 1.850

Stress factor from Table 4.16.4 [K]:

= A / B
 = 1020.0/828.0
 = 1.232

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169(K² • log10(K)) / (K² - 1)) / (K - 1)
 = (0.66845 + (5.7169(1.232² • log10(1.232)) / (1.232² - 1)) / (1.232 - 1)
 = 9.430

Stress factor from Table 4.16.4 [T]:

= (K²(1 + 8.55246 • log10(K)) - 1) / ((1.0472 + 1.9448 • K²) • (K - 1))
 = (1.232²(1 + 8.55246 • log10(1.232)) - 1) / ((1.0472 + 1.9448 • 1.232²) • (1.232 - 1))
 = 1.826

Stress factor from Table 4.16.4 [U]:

= (K²(1 + 8.55246 • log10(K)) - 1) / (1.36136(K² - 1) • (K - 1))
 = (1.232²(1 + 8.55246 • log10(1.232)) - 1) / (1.36136(1.232² - 1) • (1.232 - 1))
 = 10.363

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
 = (1.232² + 1) / (1.232² - 1)
 = 4.864

Stress factor from Table 4.16.4 for integral flanges [d]:

= U • g0² • h0 / V
 = 10.363 • 14.0² • 107.666 / 0.373
 = 585573.750 mm.³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
 = 0.893 / 107.666
 = 0.008 1/mm.

Stress factor from Table 4.16.4 [L]:

= (t • e + 1) / T + t³/d
 = (60.0 • 0.008 + 1) / 1.826 + 60.0³/585574
 = 1.189

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f • Mop / Bcor) / (L • g1²)
 = (1.8497•4565/828.0)/(1.189•24.0²)
 = 146.04 N./mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.8497 \cdot 5560 / 828.0) / (1.189 \cdot 24.0^2)$$

$$= 177.84 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.6635 \cdot 4565 / 828.0) / (1.189 \cdot 60.0^2)$$

$$= 21.01 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (1.6635 \cdot 5560 / 828.0) / (1.189 \cdot 60.0^2)$$

$$= 25.59 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (9.4301 \cdot 4565 / (60.0^2 \cdot 828.0)) - 4.8644 \cdot 21$$

$$= 39.42 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (9.4301 \cdot 5560 / (60.0^2 \cdot 828.0)) - 4.8644 \cdot 26$$

$$= 48.01 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (146 + \max(21, 39)) / 2$$

$$= 92.73 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (178 + \max(26, 48)) / 2$$

$$= 112.92 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= \text{Wm1} / \text{Ab}$$

$$= 90904 / 85.316$$

$$= 104.49 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (123905 / 85.316)$$

$$= 142.43 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	146.04	207.00	177.84	207.00
Radial Flange	21.01	138.00	25.59	138.00
Tangential Flange	39.42	138.00	48.01	138.00
Maximum Average	92.73	138.00	112.92	138.00
Bolting	104.49	172.00	142.43	172.00

Minimum Required Flange Thickness 53.543 mm.

Estimated M.A.W.P. (Operating) 18 bars
 Estimated Finished Weight of Flange at given Thk. 139.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 181.4 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot \text{Ma} / \text{Bsc} \cdot \text{Cnvfac} \cdot V / (\lambda \cdot \text{Eamb} \cdot \text{go}^2 \cdot \text{ho} \cdot \text{Ki})$$



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$$= 52.14 \cdot 5729.0 / 1.0901 \cdot 9806.637 \cdot 0.373 / (1.053 \cdot 202713 \cdot 14.0^2 \cdot 107.666 \cdot 0.3)$$

$$= 0.742 \quad (\text{should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 4704.6 / 1.0901 \cdot 9806.637 \cdot 0.373 / (1.053 \cdot 195474 \cdot 14.0^2 \cdot 107.666 \cdot 0.3)$$

$$= 0.632 \quad (\text{should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 5559.6 / 1.0579 \cdot 9806.637 \cdot 0.373 / (1.189 \cdot 202713 \cdot 14.0^2 \cdot 107.666 \cdot 0.3)$$

$$= 0.657 \quad (\text{should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 4565.5 / 1.0579 \cdot 9806.637 \cdot 0.373 / (1.189 \cdot 195474 \cdot 14.0^2 \cdot 107.666 \cdot 0.3)$$

$$= 0.560 \quad (\text{should be } \leq 1)$$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.371, Temperature Reduction per Fig. UCS 66.1 = 63 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	146.04	207.00	177.84	207.00
Radial Flange	21.01	138.00	25.59	138.00
Tangential Flange	39.42	138.00	48.01	138.00
Maximum Average	92.73	138.00	112.92	138.00
Bolting	104.49	172.00	142.43	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 53.543 mm.

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 Internal Pressure Calculations: Step: 5 9:55am Aug 12,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Chnl. head		5.5947	12	3.2	828	138	1
Chnl. rear		5.5947	12	3.2	828	138	2
Chnl. rear flang		5.516	85	3.2	828	138	3
Shell flange		12.653	68	...	828	138	4
Shell		12.653	14	3.2	828	138	5
shell head		12.653	12	3.2	828	138	6

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Chnl. head		5.516	28.6636	39.8814	12	5.7	1
Chnl. rear		5.516	28.6636	39.3132	12	5.7	2
Chnl. rear flang		5.516	17.4358	20.8273	77	72.1106	3
Shell flange		12.653	17.9347	17.9347	60	53.5432	4
Shell		12.653	35.1744	45.7352	14	7.04682	5
shell head		12.653	23.3491	33.9139	10.2	6.99045	6

Warning: Channel Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:
 For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 13 bars
 Shell Side MAPnc = 18 bars
 Channel Side MAWP = 13 bars
 Channel Side MAPnc = 20 bars

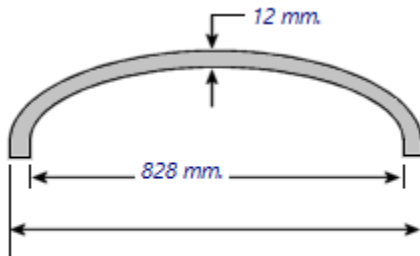
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Chnl. head

Elliptical Head Design Information:			
Design Pressure P	5.595 bars	Int. Design Temperature	71 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	50.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	828.000 mm.	Actual Stress	26.3 N./mm ²
Spec. Min. Thk t,e, or sa	12.0000 mm.	Spec. Nominal Thk.	12.000 mm.
Surface Area	9207.0 cm ²	SG of Contents	1.000
Element Volume	101248.0 cm ³	Weight of Contents	100.8 Kg
Empty Weight	90.8 Kg	Operating Weight	191.7 Kg
		Comp. All. Ope. Stress	130.9 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff. E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.350 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.175 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (5.595 \cdot 834.4 \cdot 0.99) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 5.595)$$

$$= 1.6751 + 3.2000 = 4.8751 \text{ mm.}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm. per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.079 bars

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 8.8) / (0.99 \cdot 834.4 + 0.2 \cdot 8.8)$$

$$= 29 - 0.1 = 29 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$



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$$= (2 \cdot 138.0 \cdot 1.0 \cdot 12.0) / (1.0 \cdot 828.0 + 0.2 \cdot 12.0)$$

$$= 40 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (5.595 \cdot (0.99 \cdot 834.4 + 0.2 \cdot 8.8)) / (2 \cdot 1.0 \cdot 8.8)$$

$$= 26.313 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (5.595 \cdot 417.2) / (138.0 \cdot 1.0 - 0.6 \cdot 5.595) + 3.2$$

$$= 4.896 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.079 bars](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.8) / (417.2 + 0.6 \cdot 8.8)$$

$$= 29 - 0.1 = 29 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (834.4 / (2 \cdot 210.2))^2) / 6$$

$$= 0.989890$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.132 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 12.0$, $t_r = 3.918$, $c = 3.2 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.445$, Temp. Reduction = 41 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12.0$, $t_r = 3.977$, $c = 3.2 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.452$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	13.079 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.452	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

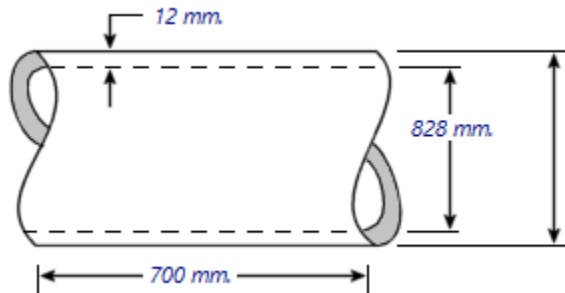
Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.350 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.175 mm.



Internal Pressure Calculation Results:

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Internal Pressure Results for: Chnl. rear

Cylindrical Shell Design Information:					
Design Pressure	P	5.595 bars	Int. Design Temperature		71 °C
Material		SA-516 70	Ext. Design Temperature		40 °C
Length		700.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.2000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	828.000 mm.	Actual Stress		26.9 N./mm ²
Spec. Min. Thk t,e, or sa		12.0000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		18736.5 cm ²	SG of Contents		1.000
Element Volume		376987.1 cm ³	Weight of Contents		372.8 Kg
Empty Weight		171.8 Kg	Operating Weight		544.6 Kg
			Comp. All. Ope. Stress		130.9 N./mm ²
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (5.595 \cdot 417.2) / (138.0 \cdot 1.0 - 0.6 \cdot 5.595) \\
 &= 1.6956 + 3.2000 = 4.8956 \text{ mm.}
 \end{aligned}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm. per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.079 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 8.8) / (417.2 + 0.6 \cdot 8.8) \\
 &= 29 - 0.1 = 29 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 12.0) / (414.0 + 0.6 \cdot 12.0) \\
 &= 39 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$



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$$= (5.595 \cdot (417.2 + 0.6 \cdot 8.8)) / (1.0 \cdot 8.8)$$
$$= 26.861 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.429 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 12.0$, $t_r = 3.977$, $c = 3.2 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.452$, Temp. Reduction = $40 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

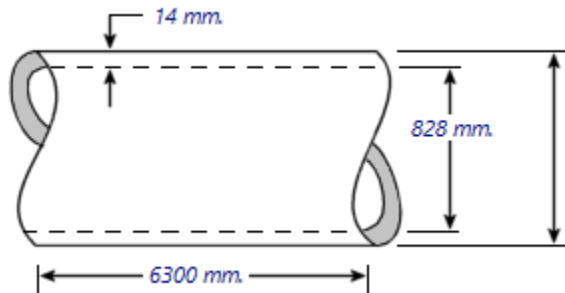
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	13.079 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	12.000 mm.	Basic MDMT	-23 °C
Stress/Thk. Ratio	0.452	Temperature Difference	25 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	12.653 bars	Int. Design Temperature	143 °C
Material	SA-516 70	Ext. Design Temperature	40 °C
Length	6300.000 mm.	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	3.2000 mm.	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm.	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	828.000 mm.	Actual Stress	49.6 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm.	Spec. Nominal Thk.	14.000 mm.
Surface Area	169419.8 cm ²	SG of Contents	0.005
Element Volume	2168555.8 cm ³	Weight of Contents	10.8 Kg
Empty Weight	1808.3 Kg	Operating Weight	1819.1 Kg
		Comp. All. Ope. Stress	135.3 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (12.653 \cdot 417.2) / (138.0 \cdot 1.0 - 0.6 \cdot 12.653) \\
 &= 3.8468 + 3.2000 = 7.0468 \text{ mm.}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 10.8) / (417.2 + 0.6 \cdot 10.8) \\
 &= 35 - 0 = 35 \text{ bars}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 14.0) / (414.0 + 0.6 \cdot 14.0) \\
 &= 46 \text{ bars}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (12.653 \cdot (417.2 + 0.6 \cdot 10.8)) / (1.0 \cdot 10.8) \\
 &= 49.642 \text{ N./mm}^2
 \end{aligned}$$



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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.663 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14.0$, $t_r = 3.964$, $c = 3.2$ mm., $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.367$, Temp. Reduction = 65 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

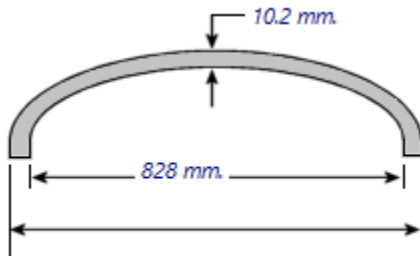
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	13.037 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	14.000 mm.	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.367	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: shell head

Elliptical Head Design Information:					
Design Pressure	P	12.653 bars	Int. Design Temperature		143 °C
Material		SA-516 70	Ext. Design Temperature		40 °C
Length		50.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02700
Int. Corr. All.	c	3.2000 mm.	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	828.000 mm.	Actual Stress		74.8 N./mm ²
Spec. Min. Thk t,e, or sa		10.2000 mm.	Spec. Nominal Thk.		12.000 mm.
Surface Area		9207.0 cm ²	SG of Contents		0.005
Element Volume		101248.0 cm ³	Weight of Contents		0.5 Kg
Empty Weight		90.8 Kg	Operating Weight		91.3 Kg
			Comp. All. Ope. Stress		135.3 N./mm ²
			Aspect Ratio		2.00

Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency		100 %	Long. Joint Eff.	E or v	100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.350 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.175 mm.

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (12.653 \cdot 834.4 \cdot 0.99) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 12.653)$$

$$= 3.7904 + 3.2000 = 6.9904 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.000 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 7.0) / (0.99 \cdot 834.4 + 0.2 \cdot 7.0)$$

$$= 23 - 0 = 23 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 10.2) / (1.0 \cdot 828.0 + 0.2 \cdot 10.2)$$

$$= 34 \text{ bars}$$



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Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (12.653 \cdot (0.99 \cdot 834.4 + 0.2 \cdot 7.0)) / (2 \cdot 1.0 \cdot 7.0)$$

$$= 74.783 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \quad \text{per UG-27 (c)(1)}$$

$$= (12.653 \cdot 417.2) / (138.0 \cdot 1.0 - 0.6 \cdot 12.653) + 3.2$$

$$= 7.047 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 0.000 bars](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \quad \text{per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 8.8) / (417.2 + 0.6 \cdot 8.8)$$

$$= 29 - 0 = 29 \text{ bars}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (834.4 / (2 \cdot 210.2)))^2 / 6$$

$$= 0.989890$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 6.132 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 10.2$, $t_r = 3.905$, $c = 3.2 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.558$, Temp. Reduction = 26 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-28 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 12.0$, $t_r = 3.964$, $c = 3.2 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.45$, Temp. Reduction = 40 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	13.037 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	11 °C
Governing Thk.	10.200 mm.	Basic MDMT	-28 °C
Stress/Thk. Ratio	0.558	Temperature Difference	20 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 10.350 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 5.175 mm.



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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	16.948 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	16.449 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	23.234 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	14.340 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	18.094 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	18.094 bars
App 27-4	= M.A.W.P.	13.037 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	17.003 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7.171 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.480 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	14.387 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	7.888 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7.888 bars
App 27-4	= M.A.W.P.	13.079 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Chnl. head	80.3	234.0	0.343	17.08
Chnl. rear	82.0	234.0	0.351	17.08
Shell	66.8	234.0	0.286	17.03
shell head	100.6	234.0	0.430	17.03

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2- Channel outl	Nozzle	118.00	118.00	1.000
N2- Channel outl	Pad	138.00	138.00	1.000
N1- Channel inle	Nozzle	118.00	118.00	1.000
N1- Channel inle	Pad	138.00	138.00	1.000
Shell inlet	Nozzle	138.00	138.00	1.000
Shell outlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Chnl. head	138.00	138.00	1.000
Chnl. rear	138.00	138.00	1.000
Chnl. rear flange-2	138.00	138.00	1.000
Shell flange	138.00	138.00	1.000
Shell	138.00	138.00	1.000
shell head	138.00	138.00	1.000

Minimum 1.000



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Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2- Channel outl	19.02	216.00	0.088
N1- Channel inle	19.02	216.00	0.088
Shell inlet	36.79	225.00	0.163
Shell outlet	18.88	225.00	0.084

Elements Suitable for Test Pressure.

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 Field Pressure Test Results: Step: 11 9:55am Aug 12,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	16.948 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	16.449 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	23.234 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	14.340 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	18.094 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	18.094 bars
App 27-4	= M.A.W.P.	13.037 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	17.003 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7.171 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.480 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	14.387 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	7.888 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	7.888 bars
App 27-4	= M.A.W.P.	13.079 bars

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 12.653 • 1.0
 = 16.449 bars

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 5.516 • 1.0
 = 7.171 bars

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Chnl. head	34.1	234.0	0.146	7.25
Chnl. rear	34.8	234.0	0.149	7.25
Shell	64.9	234.0	0.277	16.53
shell head	97.7	234.0	0.418	16.53

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N2- Channel outl	Nozzle	118.00	118.00	1.000
N2- Channel outl	Pad	138.00	138.00	1.000
N1- Channel inle	Nozzle	118.00	118.00	1.000
N1- Channel inle	Pad	138.00	138.00	1.000
Shell inlet	Nozzle	138.00	138.00	1.000
Shell outlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
-------------	---------	-----------	-------



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Chnl. head	138.00	138.00	1.000
Chnl. rear	138.00	138.00	1.000
Chnl. rear flange-2	138.00	138.00	1.000
Shell flange	138.00	138.00	1.000
Shell	138.00	138.00	1.000
shell head	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N2- Channel outl	8.07	216.00	0.037
N1- Channel inle	8.07	216.00	0.037
Shell inlet	35.71	225.00	0.159
Shell outlet	18.32	225.00	0.081

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		16.45 bars
Shell Thickness	ts	14.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	828.000 mm.
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Test Pressure		7.17 bars
Channel Thickness	tc	12.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	828.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	370
Tube Wall Thickness	et	2.5200 mm.
Tube Outside Diameter	D	25.4000 mm.
Total Straight Tube Length	Lt	6096.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		5996.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm.
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.5200 mm.
Groove Weld Leg	ag	2.5200 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	392.931 mm.
Largest Center-to-Center Tube Distance	U1	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	8.0000 mm.
Area of Tube Layout	Ap	3240.3 cm ²
Perimeter of Tube Layout	Cp	2271.6 mm.



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Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03017
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	100.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm.
Tubesheet Outside Diameter	A	1020.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	894.00 mm.
Step 1 Depth on the Tubeside	ht1	8.00 mm.
Step 2 Diameter on the Tubeside	dt2	825.00 mm.
Step 2 Depth on the Tubeside	ht2	8.00 mm.
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	894.00 mm.
Step 1 Depth on the Shellside	hs1	8.00 mm.
Step 2 Diameter on the Shellside	ds2	825.00 mm.
Step 2 Depth on the Shellside	hs2	8.00 mm.
Calculated Tubesheet Diameter per Div 2, 4.18.4		894.00 mm.

Note: Tubesheet diameter is now: 894.000 mm. Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	873.151 mm.
Area of the Untubed Lanes	AL	0.0 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	1020.000 mm.
Flange Inside Diameter	B	828.000 mm.
Flange Face Outside Diameter	Fod	897.000 mm.
Flange Face Inside Diameter	Fid	828.000 mm.
Gasket Outside Diameter	Go	886.000 mm.
Gasket Inside Diameter	Gi	860.000 mm.
Small end Hub thk.	g0	16.0000 mm.
Large end Hub thk.	g1	24.0000 mm.
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	4.5000 mm.
Full face Gasket Flange Option		Program Selects

Bolting Information:

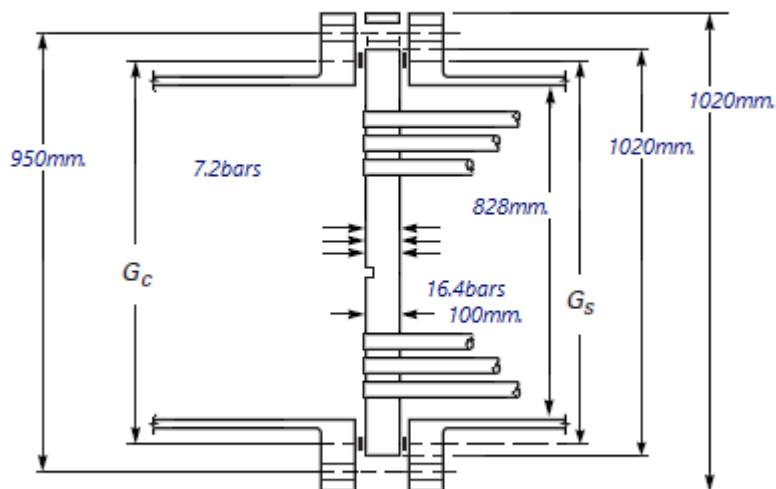
Diameter of Bolt Circle	C	950.000 mm.
Nominal Bolt Diameter	dB	25.4000 mm.
Thread Series	TEMA Thread Series	
Number of Bolts	n	24
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²



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Weld between Flange and Shell/Channel	0.0000 mm.
Alternate Flange Operating Bolt Load, Wm1	118170.03 KgF
Alternate Flange Seating Bolt Load, Wm2	123904.77 KgF
Alternate Flange Design Bolt Load, W	136768.00 KgF
Tubesheet Integral with	None
Tubesheet Extended as Flange	Yes
Thickness of Extended Portion of Tubesheet Tf	68.0000 mm.
Is Bolt Load Transferred to the Tubesheet	No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):



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Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (7.17 \cdot (25.4/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 7.17)$
 $= 0.0983 + 0.0000 = 0.0983 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 21.11 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.5200	25.40	6388.93	10.08	50.0000	0.0108275	95.84

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 95.8405) / (3 \cdot 10.0794) = 126.7738 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5686	25.40	6388.93	44.67	50.0000	0.0005513	55.11

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 55.1135) / (3 \cdot 44.6703) = 16.4495 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5686 mm.
Allowable Internal Pressure at Corroded thickness	199.14 bars
Required Internal Design Pressure	7.17 bars
Allowable External Pressure at Corroded thickness	126.77 bars
Required External Design Pressure	16.45 bars
Required Thickness due to Shell Side pressure	0.5686 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm.
Basic Gasket Width,	$b0 = N / 2$	6.500 mm.
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm.
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	873.156 mm.

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 392.931 + 25.4$
 $= 811.262 \text{ mm.}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (31.75 - 25.4) / 31.75$
 $= 0.200$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Gs / Do$ (Configurations d, e, f)
 $= 873.1555 / 811.262$
 $= 1.0763$



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Ratio [pc]:
 = G_c / D_o (Configurations d)
 = $873.1509 / 811.262$
 = 1.0763

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:
 = l_{tx} / h
 = $0 / 100.0$
 = 0.000 (must be $0 \leq p \leq 1$)

Effective tube hole Diameter [d*]:
 = $\max(d_t - 2t_t(E_t/E) \cdot (S_t/S) \cdot (p); d_t - 2t_t)$
 = $\max(25.4 - 2 \cdot 2.52 (0.20270E+09 / .20270E+09) \cdot (92/137) \cdot (0); 25.4 - 2 \cdot 2.52)$
 = 25.4000 mm.

Effective tube pitch [p*]:
 = $p / \sqrt{ 1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) }$
 = $31.75 / \sqrt{ 1 - 4 \cdot \min(0; 4 \cdot 811.262 \cdot 31.75) / (\pi \cdot 811.262^2) }$
 = 31.7500 mm.

Effective ligament efficiency for bending [μ^*]:
 = $(p^* - d^*) / p^*$
 = $(31.75 - 25.4) / 31.75$
 = 0.2000

$h/p = 3.149606$; $\mu^* = 0.200000$; $E^* = 31489462$ KPa.
 $E^*/E = 0.155349$; $v^* = 0.458572$

Note: As h/p (3.150) is > 2 , data values for $h/p = 2$ were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:
 = A / D_o
 = $894.0 / 811.262$
 = 1.1020

Coefficient [F]:
 = $(1 - v^*) / E^* \cdot (E \cdot \ln(K))$
 = $(1 - 0.46) / 31489462 \cdot (0.20270E+09 \cdot \ln(1.1))$
 = 0.3385

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:
 = $D_o^2 / 16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t]$
 = $811.262^2 / 16 \cdot [(1.076 - 1) \cdot (1.076^2 + 1) \cdot 16.449 - (1.076 - 1) \cdot (1.076^2 + 1) \cdot 7.171]$
 = 62850.0547 bars•mm.²

Moment acting on unperforated tubesheet rim [M*]
 = $M_{ts} + W^* \cdot (G_c - G_s) / (2 \cdot \pi \cdot D_o)$
 = $62850.1 + 118170.0 \cdot (873.151 - 873.151) / (2 \cdot \pi \cdot 811.262)$
 = 62850.0547 bars•mm.²

Note: W^* is the maximum of the bolt loads between the shell and channel sides.

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:



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$$= ((M \cdot) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((62850.05) - 811.262^2/32 \cdot 0.338 (16.45 - 7.17)) / (1 + 0.34)$$

$$= -1298.0842 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt)$$

$$= -1298.08 + 811.262^2/64 \cdot (3 + 0.459) \cdot (16.45 - 7.17)$$

$$= 328689.8438 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| -1298.084 | ; | 328689.844 |)$$

$$= 328689.8438 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 328689.844 / (0.2 (100.0 - 8.0)^2)$$

$$= 116.5084 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot Sy$$

$$= 2 \cdot 0.95 \cdot 250.0$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 328801.0 / (0.2 (53.5957 - 8.0)^2)$$

$$= 474.4961 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 53.5957 + 0 + 0$$

$$= 53.5957 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(53.5957 ; 3.4831)$$

$$= 53.5957 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| Ps - Pt |$]:

$$= | Ps - Pt |$$

$$= | 16.45 - 7.17 |$$

$$= 9.278 \text{ bars}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot 0.8 \cdot \beta T \cdot Sy$$

$$= 4 \cdot 0.2 \cdot 100.0 / 811.26 \cdot 0.8 \cdot 0.95 \cdot 249.995$$

$$= 18.736 \text{ N./mm}^2$$

$| Ps - Pt |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 189.997) \cdot (4 \cdot 3240.3 / 2271.59) \cdot | 16.45 - 7.17 | + 0$$

$$= 3.4831 \text{ mm.}$$

Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:

$$= \pi (do - t) t$$



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$$= \pi (25.4 - 2.52) 2.52$$

$$= 1.811 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.52 (25.4 - 2.52) 170.997$$

$$= 3158.392 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.52 (25.4 + 0.67 \cdot 2.52) 171.0$$

$$= 3158.3923 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min (S \text{ and } S_t)$$

$$= 170.997 / 170.997$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.52 (25.4 + 0.67 \cdot 2.52) 171.0$$

$$= 2056.6296 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.52 \cdot (25.4 - 2.52) 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$$

$$= 3.111 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.5557 mm.
 Minimum Required Groove Weld Leg [agr]: 1.5557 mm.

Tube-Tubesheet Jt allowable, 3158.39 is >= tube strength 3158.39 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	116.5 <=	475.0 N./mm ²	Ok
Tubesheet shear stress	6.6 <=	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	53.596	100.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.556	2.520	Ok
Tube-Tubesheet Groove Weld Leg :	1.556	2.520	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	48.103 1.397	90	475	5 190	Ps+Pt D1	Ok



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D2uc	68.675	3.204	207	475	12	190	Ps+Pt D2	Ok
D3uc	53.596	1.807	117	475	7	190	Ps+Pt D3	Ok
Max:	68.6746	3.204	mm.	0.435		0.062	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	68.675	100.000	Ok
Tubesheet thickness flanged extension :	3.204	68.000	Ok
Tube thickness :	0.569	2.520	Ok
Tube-tubesheet fillet weld leg :	1.556	2.520	Ok
Tube-tubesheet groove weld leg :	1.556	2.520	Ok

Note: This is a full strength Tube to Tubesheet Joint.

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 External Pressure Calculations: Step: 12 9:55am Aug 12,2026

External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	852	8.8	0.0014345	95.6204
20	30	819	852	8.8	0.0014777	96.3995
30	40	No Calc	...	73.8	No Calc	No Calc
40	50	No Calc	...	60	No Calc	No Calc
50	60	6419	856	10.8	0.00022069	22.0637
60	70	No Calc	848.4	7	0.0011459	89.3781

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	12	No Calc	...	10.973
20	30	12	No Calc	...	13.2749
30	40	77	72.1106	...	No Calc
40	50	60	53.5432	...	No Calc
50	60	14	No Calc	...	3.71143
60	70	10.2	No Calc	...	8.19334

Minimum

4

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	819	No Calc	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	6419	No Calc	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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 Element and Detail Weights: Step: 13 9:55am Aug 12,2026

Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	90.8154	101248	66.598	103403	9.08154
20	30	171.806	376987	126.471	382837	17.1806
30	40	189.129	56181.3	182.458	56331.8	18.9129
40	50	156.781	49546.9	156.781	49546.9	15.6781
50	60	1808.26	2168555	1400.25	2221208	180.826
60	70	90.8154	101248	66.598	103403	9.08154
Total		2507	2853767.00	1999	2916729.75	250



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset Dtl. Cent. mm.	Description
10	Liqd	100.847	-69	14	...	Cooling water
20	Liqd	372.82	350	14	...	Cooling water
20	Nozl	43.838	300	523.537	...	N2- Channel outl
20	Nozl	43.838	300	-523.537	...	N1- Channel inle
20	Wght	70	...	...	...	Partition plate
50	Sadl	102.473	1260	524	...	Left Saddle
50	Sadl	102.473	5040	524	...	Right saddle
50	Liqd	10.8411	3150	14	...	Hydrogen
50	Insl	52.3715	3150	...	...	PP
50	Nozl	56.5217	500	523.537	...	Shell inlet
50	Nozl	98.7425	5900	-550.525	...	Shell outlet
60	Liqd	0.50447	119	14	...	Hydrogen
60	Insl	3.5952	112.5	...	...	Hot insulation
30	FTsh	536.809	149.5	...	...	Fixed tubesheet
30	Tube	3716.65	3147.5	...	...	Tubes
40	Bafl	11.9409	904.81	...	...	Baffles
40	Bafl	11.9409	1909.41	...	...	Baffles
40	Bafl	11.9409	2914.01	...	...	Baffles
40	Bafl	11.9409	3918.61	...	...	Baffles
40	Bafl	11.9409	4923.21	...	...	Baffles
40	Bafl	11.9409	5927.81	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	204.9
Liquid	485.0
Insulation	56.0
Nozzles	242.9
Weights	70.0
Exchanger Components	4325.1
Liquid in Tubes	808.3
Sum of the Detail Weights	6192.3 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
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Element and Detail Weights: Step: 13 9:55am Aug 12,2026

Main Elements	2758.4	2758.4	2758.4	2758.4	2758.4	2758.4
Saddles	204.9	204.9	204.9	204.9	204.9	204.9
Nozzles	242.9	242.9	242.9	242.9	242.9	242.9
Wld Weights	70.0	70.0	70.0	70.0	70.0	70.0
Exchanger	4325.1	4325.1	4325.1	4325.1	4325.1	4325.1
Insulation	...	...	...	...	56.0	56.0
Ope. Liquid	...	...	...	...	...	485.0
Tube Ope Lqd	...	...	...	...	...	808.3
Test Liquid	...	2852.0	...	...	...	...
Tube Tst Lqd	...	808.3	...	...	...	...
Totals	7601.4	11261.7	7601.4	7601.4	7657.3	8950.7

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	7601.4 kg.
Shop Test Wt.	- Fabricated weight + water (full)	11261.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	7601.4 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	7601.4 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	7657.3 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	8950.7 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	8391.4 kg.
Field Test Wt.	- Empty weight + test medium (full)	10821.3 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	808826.7 cm ³
Weight of Ope Liq in tubes :	808.3 kg.
Weight of Water in tubes :	808.3 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	9207.01	8732.25
20	30	18736.5	18208.7
30	40	5999.06	2679.28
40	50	5626.22	2393.14
50	60	169420	163878
60	70	9207.01	8732.25

Total	218195.6	204623.7 cm ²
Total	234.9	220.3 ft ²

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 FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 56 of 170
 Nozzle Flange MAWP: Step: 14 9:55am Aug 12,2026

Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N2- Channel ou	18.51	19.60	71	150	GR 1.1	27.642	13.079	18.509	0.122
N1- Channel in	18.51	19.60	71	150	GR 1.1	27.642	13.079	18.509	0.122
Shell inlet	45.31	51.00	143	300	GR 1.1	27.642	40.329	45.314	40.329
Shell outlet	45.31	51.00	143	300	GR 1.1	23.949	44.022	45.314	44.022

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 40.329 bars
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 51.000 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 13.079 bars
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 19.600 bars

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm.	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD bars	Maximum Pressure Pmax bars	Pass Fail
N2- Channel outl	248.539	1.20	816	551	13.079	13.079	Pass
N1- Channel inle	248.539	1.20	816	551	13.079	13.079	Pass
Shell inlet	248.539	0.50	816	551	13.037	40.329	Pass
Shell outlet	302.387	0.50	1274	683	13.037	44.022	Pass

Evaluating Flange on Nozzle [N2- Channel outl]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(816) + 4 \cdot \text{abs}(551) \cdot 248.539 \leq \pi \cdot 248.539^3 [(18.509 - 13.079) + 1.200 \cdot 18.509]$$

$$= 0.13596E+05 \leq 0.13596E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N2- Channel outl [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(18.509, 1.200 \cdot 18.509 + 18.509 - 13595.7 / (\pi \cdot 248.539^3))$$

$$= \min(18.509, 13.079)$$

$$= 13.079 \text{ bars}$$

Evaluating Flange on Nozzle [N1- Channel inle]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(816) + 4 \cdot \text{abs}(551) \cdot 248.539 \leq \pi \cdot 248.539^3 [(18.509 - 13.079) + 1.200 \cdot 18.509]$$

$$= 0.13596E+05 \leq 0.13596E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1- Channel inle [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(18.509, 1.200 \cdot 18.509 + 18.509 - 13595.7 / (\pi \cdot 248.539^3))$$

$$= \min(18.509, 13.079)$$

$$= 13.079 \text{ bars}$$

Evaluating Flange on Nozzle [Shell inlet]:



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 Nozzle Flange MAWP: Step: 14 9:55am Aug 12,2026

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR] \\
 &= 16(816) + 4 \cdot \text{abs}(551) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot [(45.314 - 13.037) + .500 \cdot 45.314] \\
 &= 0.13596E+05 \leq 0.27020E+05 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle Shell inlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(45.314, 0.500 \cdot 45.314 + 45.314 - 13595.7 / (\pi \cdot 248.539^3)) \\
 &= \min(45.314, 40.329) \\
 &= 40.329 \text{ bars}
 \end{aligned}$$

Evaluating Flange on Nozzle [Shell outlet]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot [(PR - PD) + FM \cdot PR] \\
 &= 16(1274) + 4 \cdot \text{abs}(683) \cdot 302.387 \leq \pi \cdot 302.387^3 \cdot [(45.314 - 13.037) + .500 \cdot 45.314] \\
 &= 0.21214E+05 \leq 0.48662E+05 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle Shell outlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(45.314, 0.500 \cdot 45.314 + 45.314 - 21214.2 / (\pi \cdot 302.387^3)) \\
 &= \min(45.314, 44.022) \\
 &= 44.022 \text{ bars}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

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 FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 58 of 170
 Wind Load Calculation: Step: 15 9:55am Aug 12,2026

Wind Load Calculation Results:

Wind Load Calculations per India Std. IS 875 (PART 3):2015, Amd. 1&2 (2020):

Input Values:

Wind Design Code	IS 875 (PART 3):2015-2020		
Basic Wind Speed for IS-875	180	Km/hr	
Wind Zone Number	5		
Base Elevation	300	mm.	
Percent Wind for Hydrotest	33.0		
Risk Factor	[k1]	1.0	
Terrain Category		2	
Equipment Class		Class A	
Topography Factor	[k3]	1.0	
Importance Factor	[k4]	1.15	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Use Gust Response Factor (Dynamic Analysis)		No	

Calculated Values:

Vessel Length to Diameter ratio: 8.609

Design Wind Speed [Vz]:
 = Basic Wind Speed • k1 • k2,1 • k3 • k4 (for the 1st element)
 = 50.0 • 1.0 • 1.0 • 1.0 • 1.15
 = 57.496 meters/s

Design Wind Pressure per 4.5.2 [pd]:
 = max(0.7 • pz, Kd • Ka • Kc • 0.6 • Vz²)
 = max(0.7 • 1983.503, 1.0 • 1.0 • 1.0 • 0.6 • 57.5²)
 = 1983.5 N/m²

Element Wind Load (operating condition):
 = Wind Area • Cf • pd
 = 0.2 • 0.872 • 1983.503
 = 342.0 N

From	Height mm.	k1	k2	k3	Vz m/sec	Cf
10	960.00	1.0000	1.0000	1.0000	57.50	0.8722
20	960.00	1.0000	1.0000	1.0000	57.50	0.8722
30	960.00	1.0000	1.0000	1.0000	57.50	0.8722
40	960.00	1.0000	1.0000	1.0000	57.50	0.8722
50	960.00	1.0000	1.0000	1.0000	57.50	0.8722
60	960.00	1.0000	1.0000	1.0000	57.50	0.8722

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm²	Elevation mm.	Pressure Kgs/m²	Force Kgf
Partition pla	0.00	960.00	202.35	0.00

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm²	Wind Pressure Kgs/m²	Element Wind Load Kgf
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Wind Load Calculation: Step: 15 9:55am Aug 12,2026

10	20	960	1022.4	1976.66	202.347	38.3596
20	30	960	1022.4	7156.8	202.347	138.887
30	40	960	993.6	943.92	202.347	18.318
40	50	960	993.6	834.624	202.347	16.197
50	60	960	1087.2	68493.6	202.347	1329.21
60	70	960	1078.08	2248.01	202.347	43.6255

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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 FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 60 of 170
 Earthquake Load Calculation: Step: 16 9:55am Aug 12,2026

Earthquake Load Calculation Results:

Seismic Analysis Results per IS 1893 (Part 4): 2024

Input Data:

Importance Factor as Entered by User	I	2.0000
Zone Number		2
Soil Type	Rocky or Hard Soil	
Response reduction factor (Force factor)	R	2.0000
Time Period of Vibration (program or user)		0.0303 Seconds
Value of Sa/g from user Spectrum Data		0.0939
Percent Damping		2.0 %
Considering MCE for load determination		No

Calculated results:

Horizontal Seismic Coefficient per 7.3.1 [Ah]:
 $= (Sa/g)_{ss} \cdot DampVal \cdot (I/R)$
 $= (0.0939 \cdot 1.0) \cdot 1.0$
 $= 0.094$

User Spectrum Data points specified for increments of Time Period

Time Period	Sa/g
0.010	0.070
0.050	0.117
0.100	0.175
0.130	0.175
0.200	0.175
0.270	0.175
0.300	0.175
0.400	0.175
0.500	0.140
0.550	0.127
0.600	0.117
0.670	0.105
0.800	0.087
1.000	0.070
1.200	0.058
1.400	0.050
1.600	0.044
1.800	0.039
2.000	0.035
2.400	0.029
0.000	0.000

Actual Time Period = 0.0303 secs
 Computed Sa/g = 0.0939

Note:

It was indicated to ignore the damping factor, assuming that the Site Spectrum specified is per the actual damping value of this vessel.
 The value of "DampVal" has been set to 1 (not used).

Earthquake Element Load, for the first Element:

$= \text{Earthquake Weight} \cdot Ah \cdot \text{Scalar}$
 $= 1093.22 \cdot 0.0939 \cdot 1.0$
 $= 102.6 \text{ Kgf}$

Vertical load per 8.4:

$= 2/3 \cdot Ah \cdot W \cdot \text{Scalar}$
 $= 2/3 \cdot 0.0939 \cdot 8746 \cdot 1.0$



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Earthquake Load Calculation: Step: 16 9:55am Aug 12,2026

= 547.2 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	414	1093.22	102.605
20	30	414	1093.22	102.605
30	40	414	1093.22	102.605
40	50	414	1093.22	102.605
50	Sad1	414	1093.22	102.605
Sad1	60	414	1093.22	102.605
50	60	414	1093.22	102.605
60	70	414	1093.22	102.605

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 17 9:55am Aug 12,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	4183.500 mm.
Center of Gravity of the Liquid	394.388 mm.
Center of Gravity of the Insulation	4393.077 mm.
Center of Gravity of the Nozzles	3301.198 mm.
Center of Gravity of the Added Weights (Operating)	50.000 mm.
Center of Gravity of the Added Weights (Empty)	50.000 mm.
Center of Gravity of the Tubesheet(s)	899.500 mm.
Center of Gravity of the Tubes	3897.500 mm.
Center of Gravity of the Baffles	4365.810 mm.
Center of Gravity of Bare Shell New and Cold	3434.067 mm.
Center of Gravity of Bare Shell Corroded	3352.450 mm.
Vessel CG in the Operating Condition	3325.329 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	3475.247 mm.
Vessel CG in the Test Condition	3518.618 mm.

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 Saddle Calcs: Bundle Pull New: Step: 18 9:55am Aug 12,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

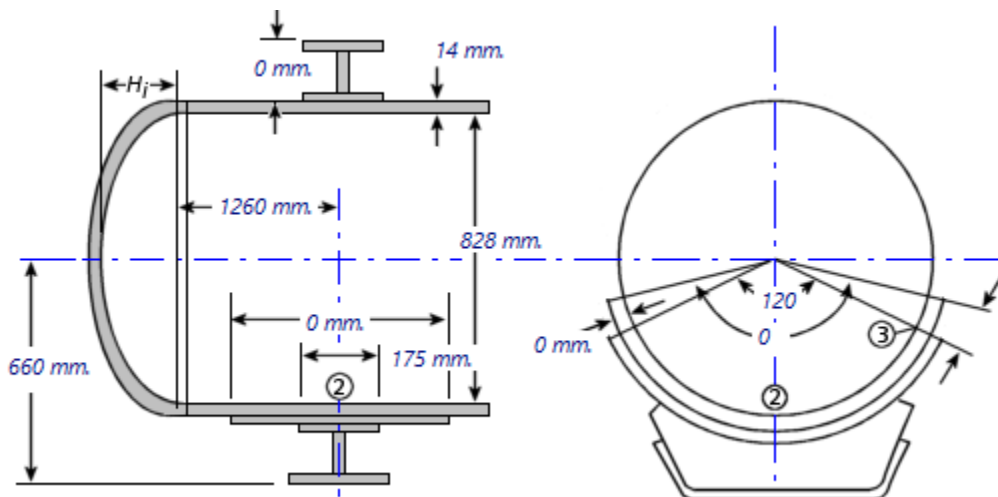
Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{421.0 \cdot 14.0}, 2 \cdot 1260.0)$
 $= 294.7650 \text{ mm.}$

Input and Calculated Values:

Vessel Mean Radius	Rm	421.00	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	6434.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm.
Saddle Width	b1 or b	175.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm.
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		4350.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		5122.46	Kgf
Pressure used in Analysis	P	0.000	bars

Horizontal Vessel Analysis Results:	Actual	Allowable
	N./mm ²	N./mm ²



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 Saddle Calcs: Bundle Pull New: Step: 18 9:55am Aug 12,2026

Long. Stress at Top of Midspan	-2.33	-234.00
Long. Stress at Top of Midspan	2.33	138.00
Long. Stress at Bottom of Midspan	2.33	138.00
Long. Stress at Top of Saddles	14.08	138.00
Long. Stress at Bottom of Saddles	-7.81	-234.00
Long. Stress at Bottom of Saddles	7.81	138.00
Tangential Shear in Shell	6.07	110.40
Circ. Stress at Horn of Saddle	23.36	207.00
Circ. Compressive Stress in Shell	0.93	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 4350 • 1.5 • 660.0/3780.0007
 = 1139.3 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 3983 + max(1139, 0, 0, 0)
 = 5122.5 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	5224.93	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	6525.00	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 720.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm^2)/3L)]$$

$$= -5122 \cdot 1260.0 [1 - (1 - 1260.0/6434.0 + (421.0^2 - 0.0^2)/(2 \cdot 1260.0 \cdot 6434.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6434.0))]$$

$$= -1193.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 5122 \cdot 6434/4 ((1 + 2(421^2 - 0^2)/(6434^2))/(1 + (4 \cdot 0)/(3 \cdot 6434)) - 4 \cdot 1260/6434)$$

$$= 1855.8 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0.0 \cdot 421.0/(2 \cdot 14.0) - 1855.8/(\pi \cdot 421.0^2 \cdot 14.0)$$

$$= -2.33 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0.0 \cdot 421.0/(2 \cdot 14.0) + 1855.8/(\pi \cdot 421.0^2 \cdot 14.0)$$

$$= 2.33 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:



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 FileName : 403-E-002ABC_Hydrogen Compressor 2nd stage inter Page 65 of 170
 Saddle Calcs: Bundle Pull New: Step: 18 9:55am Aug 12,2026

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 421.0 / (2 \cdot 14.0) - 1193.4 / (0.1066 \cdot \pi \cdot 421.0^2 \cdot 14.0)$$

$$= 14.08 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 421.0 / (2 \cdot 14.0) + 1193.4 / (0.1923 \cdot \pi \cdot 421.0^2 \cdot 14.0)$$

$$= -7.81 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 5122(6434.0 - 2 \cdot 1260.0) / (6434.0 + (4 \cdot 0.0 / 3))$$

$$= 3116.2 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 3116.15 / (421.0 \cdot 14.0)$$

$$= 6.07 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(421.0 \cdot 14.0)}$$

$$= 59.882 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$$

$$= -0.7603 \cdot 5122 \cdot 0.1 / (14.0(175.0 + 59.88 + 59.88))$$

$$= -0.93 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$$

$$= -5122 / (4 \cdot 14.0(175.0 + 59.882 + 59.882)) -$$

$$3 \cdot 0.0529 \cdot 5122 / (2 \cdot 14.0^2)$$

$$= -23.36 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(175.0 + 1.56 \cdot \sqrt{(421.0 \cdot 14.0)}, 2 \cdot 1260.0)$$

$$= 294.76 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(414.0 \cdot 14.0)} + 225.0)14.0^3) / 12$$

$$= 7.861 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 7.861 + 48.127(81.711 - 7.0)^2$$

$$= 2694.178 \text{ cm}^4$$

B

D

Y

A

AY

I + AD⁽²⁾



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	mm.	mm.	mm.	cm ²	mm. ³	cm ⁴
Shell	343.8	14.0	7.0	48.1	33689.0	0.269E+04
Wearplate	225.0	12.0	20.0	27.0	54000.0	0.103E+04
Web	16.0	204.0	128.0	32.6	417792.0	0.183E+04
BasePlate	150.0	16.0	238.0	24.0	571199.9	0.587E+04
Totals	...	...	...	131.8	1076681.0	0.114E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1076681/131.767$$

$$= 81.711 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 5122.462$$

$$= 1042.5320 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.2224 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 660.0 - 414.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 317.625 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 331.1405 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.3226 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(5122 + 102)150.0) / (2 \cdot 720.0 \cdot 173.33)}$$

$$= 9.612 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (414.0 + 14.0 + 12.0)$$

$$= 762.102 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 762.1024 / (3 - 1)$$

$$= 381.051 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 381.0512 \cdot 150.0/2$$

$$= 285.788 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$



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$$= 5224.935 / (720.0 \cdot 150.0)$$

$$= 4.838 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 285.8 \cdot 4.84$$

$$= 1382.616 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.44 + 30.484$$

$$= 55.924 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 1382.6 / 55.9241$$

$$= 2.425 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.111E+04
Web	190.5	16.0	8.0	30.5	24387.3	489.
Totals	...	...	...	55.9	267339.2	0.160E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 175.0 - 16.0$$

$$= 159.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 267339.25 / 55.924$$

$$= 47.804 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(47.804, 175.0 - 47.804)$$

$$= 127.196 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(1604.2 / 55.924)}$$

$$= 53.558 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 660.0 - \cos(120.0 / 2) (414.0 + 14.0 + 12.0) - 16.0$$

$$= 424.000 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0 / 2) (414.0 + 14.0 + 12.0)$$

$$= 762.102 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 720.0 - 50.8$$

$$= 669.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$



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$$= (762.102 - 669.2) / 2$$

$$= 46.451 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{424.0^2 + 46.451^2}$$

$$= 426.537 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 426.537 / 53.558$$

$$= 7.964$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 6525.0 / (2 \cdot 720.0) \cdot 381.051 \cdot 426.54 / 2$$

$$= 368.245 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($7.964 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.96)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (7.96) / (8 \cdot 123.21) - (7.96^3) / (8 \cdot 123.21^3))$$

$$= 153.4 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 2.42 / 153.44 + (368.24 \cdot 127.196 / 16041745) / 173.33$$

$$= 0.181$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.151E+04
Web	381.1	16.0	8.0	61.0	48774.6	418.
Totals	...	...	...	86.4	291726.5	0.192E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 291726.531 / 86.408$$

$$= 33.761 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(33.761, \text{abs}(175.0 - 33.761))$$

$$= 141.239 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 660.0 - \cos(440.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 204.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1923.3 / 86.408}$$

$$= 47.178 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 204.0 / 47.178$$

$$= 4.324$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 6525.0 / (2 \cdot 720.0)$$

$$= 4.531 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.531 \cdot 381.051 \cdot 204.0$$

$$= 352.241 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 352.241 / 136171.297$$

$$= 25.367 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($4.324 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.32)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (4.32) / (8 \cdot 123.21) - (4.32^3) / (8 \cdot 123.21^3))$$

$$= 154.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 3.46 / 154.68 + (352.24 \cdot 141.239 / 1923.262) / 173.33$$

$$= 0.169$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.0 mm.
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	5224.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	6525.0 Kg
Maximum Transverse Force	Ft	0.0 Kg
Maximum Longitudinal Force	F1	6525.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6525.0 / (4.235 \cdot 2)$$

$$= 75.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0.0 / (4.235 \cdot 2)$$

$$= 0.0 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (Q_L - Q_0) / (Stba \cdot R_{nbolts})$$

$$= (6525.0 - 5224.93) / (172.0 \cdot 2.0)$$

$$= 0.3706 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= F / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$



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$$= 6525.0 / (103.2 \cdot 2.0)$$
$$= 3.1003 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $660.0 \cdot 0.0 + 0.0$
= 0.00 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $0.0 / 5224.93$
= 0.00 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.3706, 3.1003, 0.0]$
= 3.1003 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

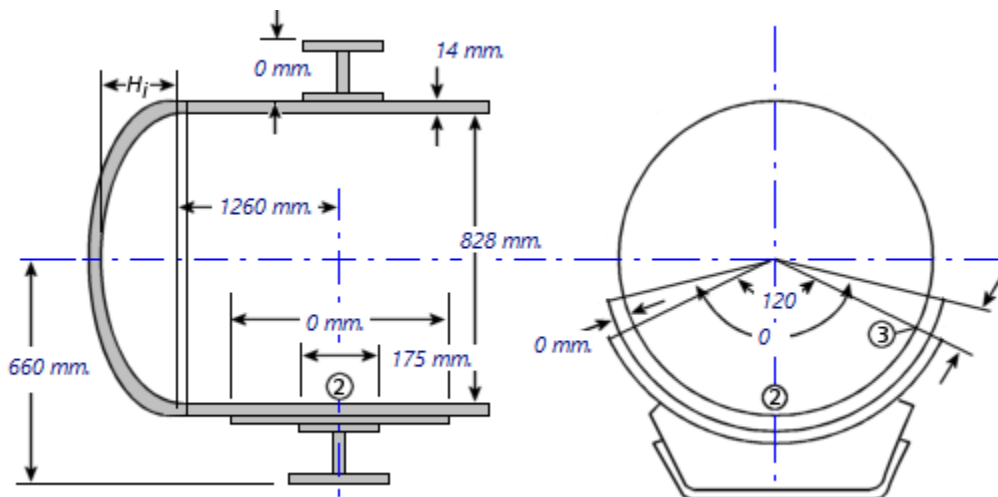
Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{422.6 \cdot 10.8}, 2 \cdot 1260.0)$
 $= 280.3904 \text{ mm.}$

Input and Calculated Values:

Vessel Mean Radius	Rm	422.60	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	6434.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm.
Saddle Width	b1 or b	175.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm.
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		4350.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force		4842.81	Kgf
Pressure used in Analysis	P	0.000	bars

Horizontal Vessel Analysis Results:	Actual	Allowable
	N./mm ²	N./mm ²



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Long. Stress at Top of Midspan	-2.84	-234.00
Long. Stress at Top of Midspan	2.84	138.00
Long. Stress at Bottom of Midspan	2.84	138.00
Long. Stress at Top of Saddles	17.12	138.00
Long. Stress at Bottom of Saddles	-9.49	-234.00
Long. Stress at Bottom of Saddles	9.49	138.00
Tangential Shear in Shell	7.41	110.40
Circ. Stress at Horn of Saddle	36.20	207.00
Circ. Compressive Stress in Shell	1.19	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 4350 • 1.5 • 660.0/3780.0007
 = 1139.3 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 3704 + max(1139, 0, 0, 0)
 = 4842.8 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4945.28	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	6525.00	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 720.000 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 = $-4843 \cdot 1260.0 [1 - (1 - 1260.0/6434.0 + (422.6^2 - 0.0^2)/(2 \cdot 1260.0 \cdot 6434.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6434.0))]$
 = -1127.8 Kg-m.

Moment per Equation 4.15.2 [M2]:
 = $Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 = $4843 \cdot 6434/4 ((1 + 2(422.6^2 - 0^2)/(6434^2))/(1 + (4 \cdot 0)/(3 \cdot 6434)) - 4 \cdot 1260/6434)$
 = 1755.0 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$
 = $0.0 \cdot 422.6/(2 \cdot 10.8) - 1755.0/(\pi \cdot 422.6^2 \cdot 10.8)$
 = -2.84 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 = $0.0 \cdot 422.6/(2 \cdot 10.8) + 1755.0/(\pi \cdot 422.6^2 \cdot 10.8)$
 = 2.84 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:



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$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 422.6 / (2 \cdot 10.8) - 1127.8 / (0.1066 \cdot \pi \cdot 422.6^2 \cdot 10.8)$$

$$= 17.12 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 422.6 / (2 \cdot 10.8) + 1127.8 / (0.1923 \cdot \pi \cdot 422.6^2 \cdot 10.8)$$

$$= -9.49 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 4843(6434.0 - 2 \cdot 1260.0) / (6434.0 + (4 \cdot 0.0 / 3))$$

$$= 2946.0 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 2946.03 / (422.6 \cdot 10.8)$$

$$= 7.41 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(422.6 \cdot 10.8)}$$

$$= 52.695 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 4843 \cdot 0.1 / (10.8(175.0 + 52.7 + 52.7))$$

$$= -1.19 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -4843 / (4 \cdot 10.8(175.0 + 52.695 + 52.695)) -$$

$$3 \cdot 0.0529 \cdot 4843 / (2 \cdot 10.8^2)$$

$$= -36.20 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(175.0 + 1.56 \cdot \sqrt{(422.6 \cdot 10.8)}, 2 \cdot 1260.0)$$

$$= 280.39 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(417.2 \cdot 10.8)} + 225.0)10.8^3) / 12$$

$$= 3.461 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 3.461 + 35.609(86.831 - 5.4)^2$$

$$= 2364.725 \text{ cm}^4$$

B

D

Y

A

AY

I + AD⁽²⁾



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	mm.	mm.	mm.	cm ²	mm. ³	cm ⁴
Shell	329.7	10.8	5.4	35.6	19229.0	0.236E+04
Wearplate	225.0	12.0	16.8	27.0	45360.0	0.133E+04
Web	16.0	204.0	124.8	32.6	407347.2	0.160E+04
BasePlate	150.0	16.0	234.8	24.0	563519.9	0.526E+04
Totals	...	...	...	119.2	1035456.1	0.106E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1035456/119.249$$

$$= 86.831 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4842.812$$

$$= 985.6173 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.1556 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 660.0 - 417.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 314.978 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 310.4543 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.5047 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(4843 + 102)150.0) / (2 \cdot 720.0 \cdot 173.33)}$$

$$= 9.351 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (414.0 + 14.0 + 12.0)$$

$$= 762.102 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 762.1024 / (3 - 1)$$

$$= 381.051 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 381.0512 \cdot 150.0/2$$

$$= 285.788 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$



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$$= 4945.285 / (720.0 \cdot 150.0)$$

$$= 4.579 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 285.8 \cdot 4.58$$

$$= 1308.616 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.44 + 30.484$$

$$= 55.924 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 1308.6 / 55.9241$$

$$= 2.295 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.111E+04
Web	190.5	16.0	8.0	30.5	24387.3	489.
Totals	...	...	...	55.9	267339.2	0.160E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 175.0 - 16.0$$

$$= 159.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 267339.25 / 55.924$$

$$= 47.804 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(47.804, 175.0 - 47.804)$$

$$= 127.196 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1604.2 / 55.924) }$$

$$= 53.558 \text{ mm.}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 660.0 - \cos(120.0 / 2) (414.0 + 14.0 + 12.0) - 16.0$$

$$= 424.000 \text{ mm.}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0 / 2) (414.0 + 14.0 + 12.0)$$

$$= 762.102 \text{ mm.}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 720.0 - 50.8$$

$$= 669.200 \text{ mm.}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$



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$$= (762.102 - 669.2) / 2$$

$$= 46.451 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{424.0^2 + 46.451^2}$$

$$= 426.537 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 426.537 / 53.558$$

$$= 7.964$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 6525.0 / (2 \cdot 720.0) \cdot 381.051 \cdot 426.54 / 2$$

$$= 368.245 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($7.964 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.96)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (7.96) / (8 \cdot 123.21) - (7.96^3) / (8 \cdot 123.21^3))$$

$$= 153.4 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 2.29 / 153.44 + (368.24 \cdot 127.196 / 16041745) / 173.33$$

$$= 0.180$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.151E+04
Web	381.1	16.0	8.0	61.0	48774.6	418.
Totals	...	...	...	86.4	291726.5	0.192E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 291726.531 / 86.408$$

$$= 33.761 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(33.761, \text{abs}(175.0 - 33.761))$$

$$= 141.239 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 660.0 - \cos(440.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 204.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{1923.3 / 86.408}$$

$$= 47.178 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 204.0 / 47.178$$

$$= 4.324$$

Unit Force [Force,u]:

$$= F / (2 \cdot \text{Baseplate Length})$$

$$= 6525.0 / (2 \cdot 720.0)$$

$$= 4.531 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.531 \cdot 381.051 \cdot 204.0$$

$$= 352.241 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 352.241 / 136171.297$$

$$= 25.367 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($4.324 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.32)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (4.32) / (8 \cdot 123.21) - (4.32^3) / (8 \cdot 123.21^3))$$

$$= 154.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$$

$$= 3.28 / 154.68 + (352.24 \cdot 141.239 / 1923.262) / 173.33$$

$$= 0.168$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.0 mm.
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	4945.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	6525.0 Kg
Maximum Transverse Force	Ft	0.0 Kg
Maximum Longitudinal Force	F1	6525.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 6525.0 / (4.235 \cdot 2)$$

$$= 75.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0.0 / (4.235 \cdot 2)$$

$$= 0.0 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (Q_L - Q_0) / (Stba \cdot R_{nbolts})$$

$$= (6525.0 - 4945.28) / (172.0 \cdot 2.0)$$

$$= 0.4504 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= F / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$



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$$= 6525.0 / (103.2 \cdot 2.0)$$
$$= 3.1003 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $660.0 \cdot 0.0 + 0.0$
= 0.00 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $0.0 / 4945.28$
= 0.00 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[Bltarearl, Bltarears, Bltareart]$
= $\max[0.4504, 3.1003, 0.0]$
= 3.1003 cm²

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

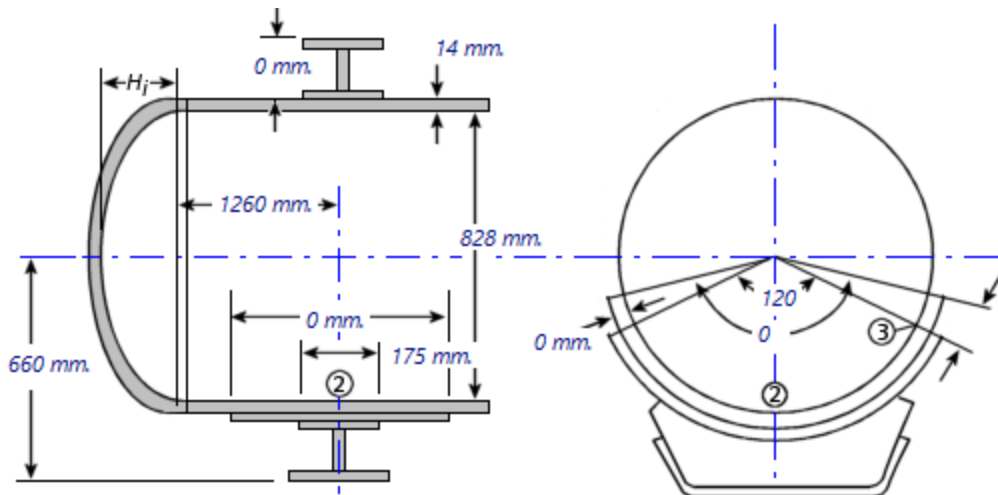
Horizontal Vessel Stress Calculations : Operating Case

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{422.6 \cdot 10.8}, 2 \cdot 1260.0)$
 $= 280.3904 \text{ mm.}$

Input and Calculated Values:

Vessel Mean Radius	Rm	422.60	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	6434.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm.
Saddle Width	b1 or b	175.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		8793.79	Kgf
Pressure used in Analysis	P	12.653	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	19.60	138.00
Long. Stress at Bottom of Midspan	29.91	138.00
Long. Stress at Top of Saddles	55.84	138.00
Long. Stress at Bottom of Saddles	7.53	138.00



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Tangential Shear in Shell	13.46	110.40
Circ. Stress at Horn of Saddle	65.73	172.50
Circ. Compressive Stress in Shell	2.17	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(1584.6/2 + 0) \cdot 660.0/720.0$
 $= 2178.8\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(184.45, 0) \cdot 660.0/3780.0007$
 $= 32.2\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(820.84, 0) \cdot 660.0/3780.0007$
 $= 143.3\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3.0(821/2 + 0) \cdot 660.0/720.0$
 $= 1128.7\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 6615 + \max(32, 2179, 143, 1129)$
 $= 8793.8\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1984.306(0.0 + \pi/4(0.856 \cdot 1.2)^2)$
 $= 1808.858\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8896.26	Kgf
Transverse Shear Load	792.30	Kgf
Longitudinal Shear Load	820.84	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= Baseplate\ Length$
 $= 720.000\ mm.$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/3L)]$
 $= -8794 \cdot 1260.0 [1 - (1 - 1260.0/6434.0 + (422.6^2 - 0.0^2) / (2 \cdot 1260.0 \cdot 6434.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 6434.0))]$
 $= -2047.9\ Kg \cdot m.$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L / 4 [(1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L]$
 $= 8794 \cdot 6434 / 4 [(1 + 2(423^2 - 0^2)/(6434^2)) / (1 + (4 \cdot 0) / (3 \cdot 6434)) - 4 \cdot 1260 / 6434]$



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$$(3 \cdot 6434) - 4 \cdot 1260 / 6434 \\ = 3186.7 \text{ Kg-m.}$$

$$\text{Longitudinal Stress at Top of Shell (4.15.4) } [\sigma_1]: \\ = P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 12.653 \cdot 422.6 / (2 \cdot 10.8) - 3186.7 / (\pi \cdot 422.6^2 \cdot 10.8) \\ = 19.60 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell (4.15.5) } [\sigma_2]: \\ = P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 12.653 \cdot 422.6 / (2 \cdot 10.8) + 3186.7 / (\pi \cdot 422.6^2 \cdot 10.8) \\ = 29.91 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Top of Shell at Support (4.15.8) } [\sigma_3]: \\ = P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 12.653 \cdot 422.6 / (2 \cdot 10.8) - 2047.9 / (0.1066 \cdot \pi \cdot 422.6^2 \cdot 10.8) \\ = 55.84 \text{ N./mm}^2$$

$$\text{Longitudinal Stress at Bottom of Shell at Support (4.15.9) } [\sigma_4]: \\ = P \cdot R_m / (2t) + M_1 / (K_1 \cdot \pi \cdot R_m^2 \cdot t) \\ = 12.653 \cdot 422.6 / (2 \cdot 10.8) + 2047.9 / (0.1923 \cdot \pi \cdot 422.6^2 \cdot 10.8) \\ = 7.53 \text{ N./mm}^2$$

$$\text{Maximum Shear Force in the Saddle (4.15.3) } [T]: \\ = Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 8794(6434.0 - 2 \cdot 1260.0) / (6434.0 + (4 \cdot 0.0 / 3)) \\ = 5349.5 \text{ Kg}$$

$$\text{Shear Stress in the shell no rings, not stiffened (4.15.12) } [\tau_2]: \\ = K_2 \cdot T / (R_m \cdot t) \\ = 1.1707 \cdot 5349.53 / (422.6 \cdot 10.8) \\ = 13.46 \text{ N./mm}^2$$

$$\text{Decay Length (4.15.20) } [x_1, x_2]: \\ = 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{422.6 \cdot 10.8} \\ = 52.695 \text{ mm.}$$

$$\text{Circumferential Stress in shell, no rings (4.15.21) } [\sigma_6]: \\ = -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2)) \\ = -0.7603 \cdot 8794 \cdot 0.1 / (10.8(175.0 + 52.7 + 52.7)) \\ = -2.17 \text{ N./mm}^2$$

$$\text{Circ. Comp. Stress at Horn of Saddle, } L \geq 8R_m \text{ (4.15.22) } [\sigma_7]: \\ = -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2) \\ = -8794 / (4 \cdot 10.8(175.0 + 52.695 + 52.695)) - \\ 3 \cdot 0.0529 \cdot 8794 / (2 \cdot 10.8^2) \\ = -65.73 \text{ N./mm}^2$$

$$\text{Effective reinforcing plate width (4.15.24) } [B_1]: \\ = \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\ = \min(175.0 + 1.56 \cdot \sqrt{422.6 \cdot 10.8}, 2 \cdot 1260.0) \\ = 280.39 \text{ mm.}$$

$$\text{Distance between Saddle Supports } [L_s]: \\ = 3780.0 \text{ mm.}$$

$$\text{Free Un-Restrained Thermal Expansion between the Saddles } [Exp]: \\ = \alpha \cdot L_s(\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.000012 \cdot 3780.001(143.3 - 21.1) \\ = 5.721 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000 mm.
Baseplate Thickness	Bpthk	16.0000 mm.



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Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12$
 $= ((1.56 \cdot \sqrt{(417.2 \cdot 10.8)} + 225.0)10.8^3)/12$
 $= 3.461 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$
 $= 3.461 + 35.609(86.831 - 5.4)^2$
 $= 2364.725 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	329.7	10.8	5.4	35.6	19229.0	0.236E+04
Wearplate	225.0	12.0	16.8	27.0	45360.0	0.133E+04
Web	16.0	204.0	124.8	32.6	407347.2	0.160E+04
BasePlate	150.0	16.0	234.8	24.0	563519.9	0.526E+04
Totals	...	...	...	119.2	1035456.1	0.106E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 1035456/119.249$
 $= 86.831 \text{ mm.}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 120.0/2$
 $= 120.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2)/(\pi - \beta + \sin(\beta)\cos(\beta))$
 $= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2)/(\pi - 2.094 + \sin(120.0)\cos(120.0))$
 $= 0.2035$

Saddle Splitting Force [Fh]:
 $= K1 \cdot Q$
 $= 0.204 \cdot 8793.785$
 $= 1789.7261 \text{ Kgf}$

Tension Stress, St = (Fh/As) = 2.0985 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 157.1972 N./mm²

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\text{theta}/2)/(\text{theta}/2 \text{ in radians})$
 $= 660.0 - 417.2 \cdot \sin(120.0/2)/1.0472$
 $= 314.978 \text{ mm.}$

Bending Moment, M = Fh * d = 563.7361 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 4.5481 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 174.6635 N./mm²

Minimum Thickness of Baseplate per Moss:



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$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(8794 + 102)150.0 / (2 \cdot 720.0 \cdot 174.663)}$$

$$= 12.494 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (414.0 + 14.0 + 12.0)$$

$$= 762.102 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 762.1024 / (3 - 1)$$

$$= 381.051 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 381.0512 \cdot 150.0 / 2$$

$$= 285.788 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 8896.258 / (720.0 \cdot 150.0)$$

$$= 8.237 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 285.8 \cdot 8.24$$

$$= 2354.118 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.44 + 30.484$$

$$= 55.924 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 2354.1 / 55.9241$$

$$= 4.128 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.111E+04
Web	190.5	16.0	8.0	30.5	24387.3	489.
Totals	...	...	...	55.9	267339.2	0.160E+04

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 175.0 - 16.0$$

$$= 159.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 267339.25 / 55.924$$

$$= 47.804 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(47.804, 175.0 - 47.804)$$

$$= 127.196 \text{ mm.}$$



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Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(1604.2/55.924)}$
 $= 53.558 \text{ mm.}$

Length of Outer Rib, step 1, [Lv]:
 $= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 660.0 - \cos(120.0/2) (414.0 + 14.0 + 12.0) - 16.0$
 $= 424.000 \text{ mm.}$

Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(120.0/2) (414.0 + 14.0 + 12.0)$
 $= 762.102 \text{ mm.}$

Length of Outer Rib, step 3, [Bd]:
 $= \text{Baseplate length} - \text{clearance}$
 $= 720.0 - 50.8$
 $= 669.200 \text{ mm.}$

Length of Outer Rib, step 4, [Dd]:
 $= (Lw - Bd) / 2$
 $= (762.102 - 669.2) / 2$
 $= 46.451 \text{ mm.}$

Length of Outer Rib [L]:
 $= \sqrt{(Lv^2 + Dd^2)}$
 $= \sqrt{(424.0^2 + 46.451^2)}$
 $= 426.537 \text{ mm.}$

Intermediate Term [Cc]:
 $= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 $= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0)}$
 $= 122.739$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 426.537/53.558$
 $= 7.964$

Bending Moment [Rm]:
 $= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$
 $= 820.8 / (2 \cdot 720.0) \cdot 381.051 \cdot 426.54/2$
 $= 46.325 \text{ Kg-m.}$

Compressive Allowable, $KL/r < Cc$ ($7.964 < 122.7394$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (7.96)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (7.96) / (8 \cdot 122.74) - (7.96^3) / (8 \cdot 122.74^3))$
 $= 154.6 \text{ N./mm}^2$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$= Sc/Sca + (Rm \cdot C1 / I) / Sba$
 $= 4.13/154.61 + (46.32 \cdot 127.196/16041745) / 174.66$
 $= 0.047$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.151E+04
Web	381.1	16.0	8.0	61.0	48774.6	418.
Totals	...	...	...	86.4	291726.5	0.192E+04



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Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 291726.531/86.408$$

$$= 33.761 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(ytot, \text{abs}(\text{Saddle Width} - ytot))$$

$$= \max(33.761, \text{abs}(175.0 - 33.761))$$

$$= 141.239 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 660.0 - \cos(440.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 204.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1923.3/86.408) }$$

$$= 47.178 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 204.0/47.178$$

$$= 4.324$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 820.839 / (2 \cdot 720.0)$$

$$= 0.570 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.57 \cdot 381.051 \cdot 204.0$$

$$= 44.312 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 44.312/136171.297$$

$$= 3.191 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($4.324 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.32)^2 / (2 \cdot 122.74^2)) 262 /$$

$$(5/3 + 3 \cdot (4.32) / (8 \cdot 122.74) - (4.32^3) / (8 \cdot 122.74^3))$$

$$= 155.9 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 5.89/155.86 + (44.31 \cdot 141.239/1923.262) / 174.66$$

$$= 0.056$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.29 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.0 mm.
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	6717.4 Kg



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Saddle Load QL (Wind/Seismic contribution)	QL	143.3 Kgf
Maximum Transverse Force	Ft	792.3 Kgf
Maximum Longitudinal Force	F1	820.8 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 834.1 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = $\max(F_f, F_1) / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $834.1 / (4.235 \cdot 2)$
 = 9.7 N./mm². Must be less than 107.9 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 = $792.3 / (4.235 \cdot 2)$
 = 9.2 N./mm². Must be less than 107.9 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = $F_1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 = $820.84 / (107.87 \cdot 2.0)$
 = 0.3731 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $660.0 \cdot 792.3 + 0.0$
 = 522.93 Kg-m.

Eccentricity (e):
 = R_{mom} / QO
 = $522.93 / 6717.44$
 = 77.84 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 0.3731, 0.0]$
 = 0.3731 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 12.0, c = 3.2 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.367$, Temp. Reduction = 65 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$
 and less than 2a. The wear plate will be ignored.



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Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 \sqrt{ (Rm \cdot t) }, 2a)$$

$$= \min(175.0 + 1.56 \sqrt{ (422.6 \cdot 10.8) }, 2 \cdot 1260.0)$$

$$= 280.3904 \text{ mm.}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	422.60	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	6434.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm.
Saddle Width	b1 or b	175.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	213.70	mm.
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		4856.80	Kgf
Pressure used in Analysis	P	12.653	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	22.48	138.00
Long. Stress at Bottom of Midspan	27.04	138.00
Long. Stress at Top of Saddles	45.39	138.00
Long. Stress at Bottom of Saddles	13.32	138.00
Tangential Shear in Shell	7.12	110.40
Circ. Stress at Horn of Saddle	36.31	172.50
Circ. Compressive Stress in Shell	1.20	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(1584.6/2 + 0) \cdot 660.0/720.0$
 $= 2178.8 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(184.45, 0) \cdot 660.0/3780.0007$
 $= 32.2 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / \text{Ls}$
 $= \max(820.84, 0) \cdot 660.0/3780.0007$
 $= 143.3 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(821/2 + 0) \cdot 660.0/720.0$
 $= 1128.7 \text{ Kgf}$



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Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 2678 + \max(32, 2179, 143, 1129)$$

$$= 4856.8 \text{ Kg}$$

Longitudinal Wind Force [F_L]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1984.306(0.0 + \pi/4(0.856 \cdot 1.2)^2)$$

$$= 1808.858 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4959.27	Kgf
Transverse Shear Load	792.30	Kgf
Longitudinal Shear Load	820.84	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 720.000 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -4857 \cdot 1260.0 [1 - (1 - 1260.0/6434.0 + (422.6^2 - 213.7^2)/(2 \cdot 1260.0 \cdot 6434.0))/(1 + (4 \cdot 213.7)/(3 \cdot 6434.0))]$$

$$= -1359.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 4857 \cdot 6434/4((1 + 2(423^2 - 214^2)/(6434^2))/(1 + (4 \cdot 214)/(3 \cdot 6434)) - 4 \cdot 1260/6434)$$

$$= 1409.4 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 12.653 \cdot 422.6/(2 \cdot 10.8) - 1409.4/(\pi \cdot 422.6^2 \cdot 10.8)$$

$$= 22.48 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 12.653 \cdot 422.6/(2 \cdot 10.8) + 1409.4/(\pi \cdot 422.6^2 \cdot 10.8)$$

$$= 27.04 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*₃]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 12.653 \cdot 422.6/(2 \cdot 10.8) - 1359.1/(0.1066 \cdot \pi \cdot 422.6^2 \cdot 10.8)$$

$$= 45.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*₄]:

$$= P \cdot Rm/(2t) + M1/(K1* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 12.653 \cdot 422.6/(2 \cdot 10.8) + 1359.1/(0.1923 \cdot \pi \cdot 422.6^2 \cdot 10.8)$$

$$= 13.32 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 4857(6434.0 - 2 \cdot 1260.0)/(6434.0 + (4 \cdot 213.7/3))$$

$$= 2829.2 \text{ Kg}$$



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K_2 \cdot T / (R_m \cdot t)$
 $= 1.1707 \cdot 2829.25 / (422.6 \cdot 10.8)$
 $= 7.12 \text{ N./mm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(R_m \cdot t)}$
 $= 0.78 \cdot \sqrt{(422.6 \cdot 10.8)}$
 $= 52.695 \text{ mm.}$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:
 $= -K_5 \cdot Q \cdot k / (t(b + X_1 + X_2))$
 $= -0.7603 \cdot 4857 \cdot 0.1 / (10.8(175.0 + 52.7 + 52.7))$
 $= -1.20 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [σ_7]:
 $= -Q / (4 \cdot t(b + X_1 + X_2)) - 3 \cdot K_7 \cdot Q / (2 \cdot t^2)$
 $= -4857 / (4 \cdot 10.8(175.0 + 52.695 + 52.695)) -$
 $3 \cdot 0.0529 \cdot 4857 / (2 \cdot 10.8^2)$
 $= -36.31 \text{ N./mm}^2$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{(422.6 \cdot 10.8)}, 2 \cdot 1260.0)$
 $= 280.39 \text{ mm.}$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	262.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$
 $= ((1.56 \cdot \sqrt{(417.2 \cdot 10.8)} + 225.0)10.8^3) / 12$
 $= 3.461 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 3.461 + 35.609(86.831 - 5.4)^2$
 $= 2364.725 \text{ cm}^4$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	329.7	10.8	5.4	35.6	19229.0	0.236E+04
Wearplate	225.0	12.0	16.8	27.0	45360.0	0.133E+04
Web	16.0	204.0	124.8	32.6	407347.2	0.160E+04
BasePlate	150.0	16.0	234.8	24.0	563519.9	0.526E+04
Totals	...	...	...	119.2	1035456.1	0.106E+05

Distance to Centroid [C1]:
 $= AY / A$
 $= 1035456 / 119.249$
 $= 86.831 \text{ mm.}$



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Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4856.802$$

$$= 988.4645 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.1590 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 157.1972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 660.0 - 417.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 314.978 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 311.3511 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.5119 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 174.6635 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(4857 + 102) \cdot 150.0) / (2 \cdot 720.0 \cdot 174.663)}$$

$$= 9.328 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (414.0 + 14.0 + 12.0)$$

$$= 762.102 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 762.1024 / (3 - 1)$$

$$= 381.051 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 381.0512 \cdot 150.0/2$$

$$= 285.788 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 4959.275 / (720.0 \cdot 150.0)$$

$$= 4.592 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 285.8 \cdot 4.59$$

$$= 1312.318 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 25.44 + 30.484$$

$$= 55.924 \text{ cm}^2$$



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Compressive Stress [Sc]:

= P/A_r
 = 1312.3/55.9241
 = 2.301 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.111E+04
Web	190.5	16.0	8.0	30.5	24387.3	489.
Totals	...	...	...	55.9	267339.2	0.160E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 175.0 - 16.0
 = 159.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 267339.25/55.924
 = 47.804 mm.

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(47.804, 175.0 - 47.804)
 = 127.196 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{1604.2/55.924}$
 = 53.558 mm.

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 660.0 - cos(120.0/2)(414.0 + 14.0 + 12.0) - 16.0
 = 424.000 mm.

Length of Outer Rib, step 2, [Lw]:

= $2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 = $2 \cdot \sin(120.0/2) (414.0 + 14.0 + 12.0)$
 = 762.102 mm.

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance
 = 720.0 - 50.8
 = 669.200 mm.

Length of Outer Rib, step 4, [Dd]:

= $(L_w - B_d) / 2$
 = $(762.102 - 669.2) / 2$
 = 46.451 mm.

Length of Outer Rib [L]:

= $\sqrt{L_v^2 + D_d^2}$
 = $\sqrt{424.0^2 + 46.451^2}$
 = 426.537 mm.

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/262.0}$
 = 122.739



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 426.537/53.558$$

$$= 7.964$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 820.8 / (2 \cdot 720.0) \cdot 381.051 \cdot 426.54 / 2$$

$$= 46.325 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($7.964 < 122.7394$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (7.96)^2 / (2 \cdot 122.74^2)) 262 / (5/3 + 3 \cdot (7.96) / (8 \cdot 122.74) - (7.96^3) / (8 \cdot 122.74^3))$$

$$= 154.6 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 2.3 / 154.61 + (46.32 \cdot 127.196 / 16041745) / 174.66$$

$$= 0.036$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.151E+04
Web	381.1	16.0	8.0	61.0	48774.6	418.
Totals	...	...	...	86.4	291726.5	0.192E+04

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 291726.531 / 86.408$$

$$= 33.761 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{abs}(\text{Saddle Width} - y_{tot}))$$

$$= \max(33.761, \text{abs}(175.0 - 33.761))$$

$$= 141.239 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - B_{pthk}$$

$$= 660.0 - \cos(440.0/2) (16.0 + 0.0 + 0.0) - 0.0$$

$$= 204.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (1923.3 / 86.408) }$$

$$= 47.178 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 204.0 / 47.178$$

$$= 4.324$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 820.839 / (2 \cdot 720.0)$$

$$= 0.570 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.57 \cdot 381.051 \cdot 204.0$$

$$= 44.312 \text{ Kg-m.}$$



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Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 44.312/136171.297

= 3.191 N./mm²

Compressive Allowable, $KL/r < C_c$ (4.324 < 122.7394) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)

= (1-(4.32)²/(2 • 122.74²))262/

(5/3+3•(4.32)/(8• 122.74)-(4.32³)/(8•122.74³)

= 155.9 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$

= 3.29/155.86 + (44.31 • 141.239/1923.262)/174.66

= 0.039

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	137.29 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.0 mm.
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	175.00 N./mm ²
Area Available in a Single Bolt	BlArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	2780.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	143.3 Kg
Maximum Transverse Force	Ft	792.3 Kg
Maximum Longitudinal Force	F1	820.8 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 820.8/(4.235 • 2)

= 9.5 N./mm². Must be less than 107.9 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= 792.3/(4.235 • 2)

= 9.2 N./mm². Must be less than 107.9 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= 820.84/(107.87 • 2.0)

= 0.3731 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= 660.0 • 792.3 + 0.0

= 522.93 Kg-m.

Eccentricity (e):

= R_{mom} / Q_0

= 522.93/2780.46



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= 188.07 mm. > Bplen/6 --> Uplift in Transverse direction

f = Bplen / 2 - Edgedis
 = 720.0/2 - 50.0
 = 310.00 mm.

Modular Ratio Of Steel/Concrete (n1):
 = ES / EC
 = 203402.5/21526.32
 = 9.45

K1 = 3 (e - 0.5 • Bplen)
 = 3 (188.07 - 0.5•720.0)
 = -515.79 mm.

K2 = 6 • n1 • At / Bpwid • (f + e)
 = 6 • 9.45 • 4.23/150.0 • (310.0 + 188.07)
 = 79723.13 mm.²

K3 = -K2 • (0.5 • Bplen + f)
 = -79723.13 • (0.5 • 720.0 + 310.0)
 = -53414502.73 mm.³

Iteratively Solving for the Effective Bearing Length:
 $Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -515.79 \cdot Y^2 + 79723.13 \cdot Y + -0.5E+08 = 0$
 Y = 548.13 mm.

Num = (Bplen / 2 - Y / 3 - e)
 = (720.0/2 - 548.13/3 - 188.07)
 = -10.78

Denom = (Bplen / 2 - Y / 3 + f)
 = (720.0/2 - 548.13/3 + 310.0)
 = 487.29

Total Bolt Tension Force [Tforce]:
 = - QO • Num / Denom
 = - 2780.46 • -10.78/487.29
 = 61.51 Kgf

Bolt Area Required due to Transverse Load [Bltareart]:
 = Tforce / (Stba • Nbt)
 = 61.51/(137.29 • 1.0)
 = 0.0439 cm²

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 0.3731, 0.0439]
 = 0.3731 cm²

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)
 = 2(QO + Tforce) / (Y • Bpwid)
 = 2(2780.46 + 61.51)/(548.13 • 150.0)
 = 6.78 bars

Distance from Baseplate Edge to the Web [ADIST]:
 = (Bplen - WebIngh) / 2
 = (720.0 - 669.2)/2
 = 25.4000 mm.

Overturning Moment due To Bolt Tension [Mt]:



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$$\begin{aligned} &= T_{force} \cdot A_{dist} \\ &= 61.51 \cdot 25.4 \\ &= 1.56 \text{ Kg-m.} \end{aligned}$$

Equivalent Bearing Pressure (f1):

$$\begin{aligned} &= f_c \cdot (Y - A_{dist}) / Y \\ &= 6.78 \cdot (548.13 - 25.4) / 548.13 \\ &= 6.47 \text{ bars} \end{aligned}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$\begin{aligned} &= (A_{dist}^2 \cdot B_{pwid} / 6) \cdot (f1 + 2 \cdot f_c) \\ &= (25.4^2 \cdot 150.0 / 6) \cdot (6.47 + 2 \cdot 6.78) \\ &= 3.29 \text{ Kg-m.} \end{aligned}$$

Baseplate Required Thickness [Treq]:

$$\begin{aligned} &= \sqrt{(6 \cdot \max(M_t, M_c) / (B_{pwid} \cdot S_{ba}))} \\ &= \sqrt{(6 \cdot \max(1.56, 3.29) / (150.0 \cdot 262.49))} \\ &= 2.2185 \text{ mm.} \end{aligned}$$

MDMT Calculations for: Right saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, $t_g = 12.0$, $c = 3.2 \text{ mm.}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.367$, Temp. Reduction = 65 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

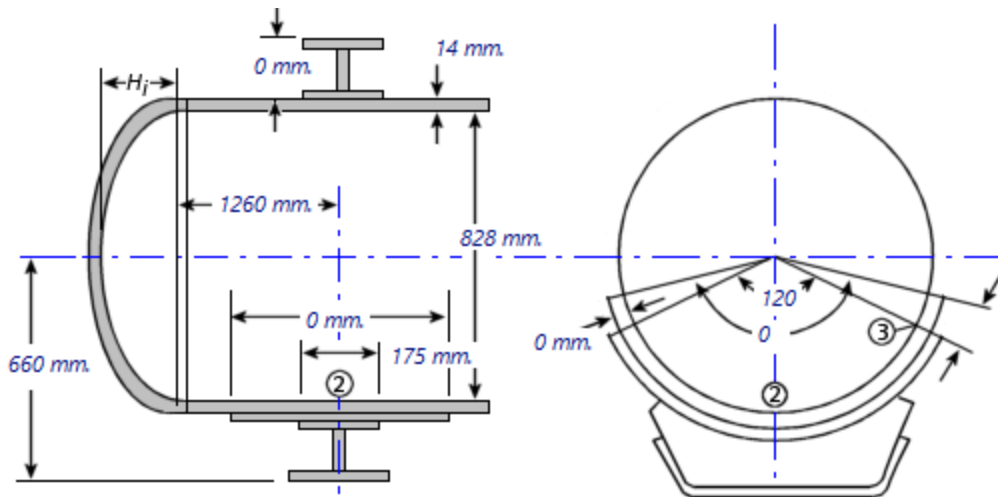
Horizontal Vessel Stress Calculations : Test Case

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$
 and less than $2a$. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{422.6 \cdot 10.8}, 2 \cdot 1260.0)$
 $= 280.3904 \text{ mm.}$

Input and Calculated Values:

Vessel Mean Radius	Rm	422.60	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	6434.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm.
Saddle Width	b1 or b	175.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm.
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		8228.27	Kgf
Pressure used in Analysis	P	16.490	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	27.44	234.00
Long. Stress at Bottom of Midspan	37.09	234.00
Long. Stress at Top of Saddles	61.35	234.00
Long. Stress at Bottom of Saddles	16.14	234.00



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Tangential Shear in Shell	12.59	138.58
Circ. Stress at Horn of Saddle	61.51	350.99
Circ. Compressive Stress in Shell	2.03	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (522.9/2 + 0) \cdot 660.0/720.0$
 $= 719.0 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(60.87, 0) \cdot 660.0/3780.0007$
 $= 10.6 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 7509 + \max(11, 719, 0, 0)$
 $= 8228.3 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.856 \cdot 1.2)^2)$
 $= 1808.858 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	8330.74	Kgf
Transverse Shear Load	261.46	Kgf
Longitudinal Shear Load	60.87	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 720.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -8228 \cdot 1260.0 [1 - (1 - 1260.0/6434.0 + (422.6^2 - 0.0^2) / (2 \cdot 1260.0 \cdot 6434.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 6434.0))]$
 $= -1916.2 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 8228 \cdot 6434/4 ((1 + 2(422.6^2 - 0^2)/(6434^2)) / (1 + (4 \cdot 0)/(3 \cdot 6434)) - 4 \cdot 1260/6434)$
 $= 2981.8 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$
 $= 16.49 \cdot 422.6/(2 \cdot 10.8) - 2981.8/(\pi \cdot 422.6^2 \cdot 10.8)$
 $= 27.44 \text{ N/mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$



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$$= 16.49 \cdot 422.6 / (2 \cdot 10.8) + 2981.8 / (\pi \cdot 422.6^2 \cdot 10.8)$$

$$= 37.09 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 t)$$

$$= 16.49 \cdot 422.6 / (2 \cdot 10.8) - 1916.2 / (0.1066 \cdot \pi \cdot 422.6^2 \cdot 10.8)$$

$$= 61.35 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 16.49 \cdot 422.6 / (2 \cdot 10.8) + 1916.2 / (0.1923 \cdot \pi \cdot 422.6^2 \cdot 10.8)$$

$$= 16.14 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 8228(6434.0 - 2 \cdot 1260.0) / (6434.0 + (4 \cdot 0.0 / 3))$$

$$= 5005.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [$\tau2$]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 5005.51 / (422.6 \cdot 10.8)$$

$$= 12.59 \text{ N./mm}^2$$

Decay Length (4.15.20) [$x1, x2$]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(422.6 \cdot 10.8)}$$

$$= 52.695 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [$\sigma6$]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 8228 \cdot 0.1 / (10.8(175.0 + 52.7 + 52.7))$$

$$= -2.03 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.22) [$\sigma7$]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -8228 / (4 \cdot 10.8(175.0 + 52.695 + 52.695)) -$$

$$3 \cdot 0.0529 \cdot 8228 / (2 \cdot 10.8^2)$$

$$= -61.51 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(175.0 + 1.56 \cdot \sqrt{(422.6 \cdot 10.8)}, 2 \cdot 1260.0)$$

$$= 280.39 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(417.2 \cdot 10.8)} + 225.0)^3) / 12$$

$$= 3.461 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$



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$$= 3.461 + 35.609(86.831 - 5.4)^2$$

$$= 2364.725 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² (2) cm ⁴
Shell	329.7	10.8	5.4	35.6	19229.0	0.236E+04
Wearplate	225.0	12.0	16.8	27.0	45360.0	0.133E+04
Web	16.0	204.0	124.8	32.6	407347.2	0.160E+04
BasePlate	150.0	16.0	234.8	24.0	563519.9	0.526E+04
Totals	...	...	...	119.2	1035456.1	0.106E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1035456/119.249$$

$$= 86.831 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 8228.27$$

$$= 1674.6312 \text{ Kg}$$

Tension Stress, $St = (Fh/As) = 1.9635 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 660.0 - 417.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 314.978 \text{ mm.}$$

Bending Moment, $M = Fh \cdot d = 527.4830 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 4.2557 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(8228 + 102)150.0) / (2 \cdot 720.0 \cdot 173.33)}$$

$$= 12.137 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (414.0 + 14.0 + 12.0)$$

$$= 762.102 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 762.1024 / (3 - 1)$$

$$= 381.051 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot B_{\text{pwid}} / 2$$

$$= 381.0512 \cdot 150.0/2$$



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$$= 285.788 \text{ cm}^2$$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 8330.742 / (720.0 \cdot 150.0)$
 $= 7.714 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 285.8 \cdot 7.71$
 $= 2204.472 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 25.44 + 30.484$
 $= 55.924 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 2204.5 / 55.9241$
 $= 3.866 \text{ N/mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.111E+04
Web	190.5	16.0	8.0	30.5	24387.3	489.
Totals	...	...	...	55.9	267339.2	0.160E+04

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 175.0 - 16.0$
 $= 159.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 267339.25 / 55.924$
 $= 47.804 \text{ mm.}$

Distance to Centroid [C1]:
 $= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$
 $= \max(47.804, 175.0 - 47.804)$
 $= 127.196 \text{ mm.}$

Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(1604.2 / 55.924)}$
 $= 53.558 \text{ mm.}$

Length of Outer Rib, step 1, [Lv]:
 $= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 660.0 - \cos(120.0/2) (414.0 + 14.0 + 12.0) - 16.0$
 $= 424.000 \text{ mm.}$

Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(120.0/2) (414.0 + 14.0 + 12.0)$
 $= 762.102 \text{ mm.}$

Length of Outer Rib, step 3, [Bd]:
 $= \text{Baseplate length} - \text{clearance}$
 $= 720.0 - 50.8$



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= 669.200 mm.

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (762.102 - 669.2)/2
 = 46.451 mm.

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$
 = $\sqrt{(424.0^2 + 46.451^2)}$
 = 426.537 mm.

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 426.537/53.558
 = 7.964

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 60.9 / (2 • 720.0) • 381.051 • 426.54/2
 = 3.435 Kg-m.

Compressive Allowable, KL/r < Cc (7.964 < 123.2105) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
 = (1 - (7.96)² / (2 • 123.21²)) 260 /
 (5/3 + 3 • (7.96) / (8 • 123.21) - (7.96³) / (8 • 123.21³))
 = 153.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 3.87/153.44 + (3.44 • 127.196/16041745) / 173.33
 = 0.027

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.151E+04
Web	381.1	16.0	8.0	61.0	48774.6	418.
Totals	...	...	...	86.4	291726.5	0.192E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 291726.531/86.408
 = 33.761 mm.

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))
 = max(33.761, abs(175.0 - 33.761))
 = 141.239 mm.

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 660.0 - cos(440.0/2) (16.0 + 0.0 + 0.0) - 0.0
 = 204.000 mm.

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$
 = $\sqrt{(1923.3/86.408)}$



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= 47.178 mm.

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 204.0/47.178
 = 4.324

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 60.87/(2 • 720.0)
 = 0.042 Kg/mm.

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 0.042 • 381.051 • 204.0
 = 3.286 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 3.286/136171.297
 = 0.237 N./mm²

Compressive Allowable, KL/r < Cc (4.324 < 123.2105) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))
 = (1-(4.32)²/(2 • 123.21²))260/
 (5/3+3•(4.32)/(8• 123.21)-(4.32³)/(8•123.21³))
 = 154.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 5.52/154.68 + (3.29 • 141.239/1923.262)/173.33
 = 0.037

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.0 mm.
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	7611.7 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	10.6 Kg
Maximum Transverse Force	Ft	261.5 Kg
Maximum Longitudinal Force	Fl	60.9 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= FI / (Bolt Area • Number of Bolts)
 = 60.9/(4.235 • 2)
 = 0.7 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= Ft / (Bolt Area • Number of Bolts)
 = 261.5/(4.235 • 2)
 = 3.0 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)



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Bolt Area due to Shear Load [Bltarears]:
 $= F / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 60.87 / (103.2 \cdot 2.0)$
 $= 0.0289 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 660.0 \cdot 261.46 + 0.0$
 $= 172.57 \text{ Kg-m.}$

Eccentricity (e):
 $= \text{Rmom} / \text{QO}$
 $= 172.57 / 7611.73$
 $= 22.67 \text{ mm.} < \text{Bplen} / 6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0.0, 0.0289, 0.0]$
 $= 0.0289 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Note:
 Wear Pad Width (225.00) is less than $1.56 \cdot \sqrt{\text{rm} \cdot t}$
 and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 $= \min(b + 1.56 \cdot \sqrt{ \text{Rm} \cdot t }, 2a)$
 $= \min(175.0 + 1.56 \cdot \sqrt{ 422.6 \cdot 10.8 }, 2 \cdot 1260.0)$
 $= 280.3904 \text{ mm.}$

Input and Calculated Values:

Vessel Mean Radius	Rm	422.60	mm.
Shell Thickness used in this Case	t	14.000	mm.
Stiffened Vessel Length per 4.15.6	L	6434.00	mm.
Distance from Saddle to Vessel tangent	a1 or a	1260.00	mm.
Saddle Width	b1 or b	175.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Inside or Mean Depth of Head	Hi or hm	213.70	mm.
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	660.00	mm.
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		4266.52	Kgf
Pressure used in Analysis	P	16.490	bars

Horizontal Vessel Analysis Results:	Actual	Allowable
	N./mm ²	N./mm ²



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Long. Stress at Top of Midspan	30.26	234.00
Long. Stress at Bottom of Midspan	34.27	234.00
Long. Stress at Top of Saddles	50.39	234.00
Long. Stress at Bottom of Saddles	22.22	234.00
Tangential Shear in Shell	6.25	138.58
Circ. Stress at Horn of Saddle	31.89	350.99
Circ. Compressive Stress in Shell	1.05	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (522.9/2 + 0) \cdot 660.0/720.0$
 $= 719.0 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(60.87, 0) \cdot 660.0/3780.0007$
 $= 10.6 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 3548 + \max(11, 719, 0, 0)$
 $= 4266.5 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1984.306 (0.0 + \pi/4 (0.856 \cdot 1.2)^2)$
 $= 1808.858 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	4368.99	Kgf
Transverse Shear Load	261.46	Kgf
Longitudinal Shear Load	60.87	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 720.000 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m^2)/3L)]$
 $= -4267 \cdot 1260.0 [1 - (1 - 1260.0/6434.0 + (422.6^2 - 213.7^2) / (2 \cdot 1260.0 \cdot 6434.0)) / (1 + (4 \cdot 213.7^2) / (3 \cdot 6434.0))]$
 $= -1193.9 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 4267 \cdot 6434/4 ((1 + 2(422.6^2 - 213.7^2)/(6434^2)) / (1 + (4 \cdot 213.7^2)/(3 \cdot 6434)) - 4 \cdot 1260/6434)$
 $= 1238.1 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 $= 16.49 \cdot 422.6 / (2 \cdot 10.8) - 1238.1 / (\pi \cdot 422.6^2 \cdot 10.8)$



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$$= 30.26 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 16.49 \cdot 422.6 / (2 \cdot 10.8) + 1238.1 / (\pi \cdot 422.6^2 \cdot 10.8)$$

$$= 34.27 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 16.49 \cdot 422.6 / (2 \cdot 10.8) - 1193.9 / (0.1066 \cdot \pi \cdot 422.6^2 \cdot 10.8)$$

$$= 50.39 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 16.49 \cdot 422.6 / (2 \cdot 10.8) + 1193.9 / (0.1923 \cdot \pi \cdot 422.6^2 \cdot 10.8)$$

$$= 22.22 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 4267(6434.0 - 2 \cdot 1260.0) / (6434.0 + (4 \cdot 213.7/3))$$

$$= 2485.4 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 2485.39 / (422.6 \cdot 10.8)$$

$$= 6.25 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{Rm \cdot t}$$

$$= 0.78 \cdot \sqrt{422.6 \cdot 10.8}$$

$$= 52.695 \text{ mm.}$$

Circumferential Stress in shell, no rings (4.15.21) [σ_6]:

$$= -K5 \cdot Q \cdot k / (t(b + X1 + X2))$$

$$= -0.7603 \cdot 4267 \cdot 0.1 / (10.8(175.0 + 52.7 + 52.7))$$

$$= -1.05 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.22) [σ_7]:

$$= -Q / (4 \cdot t(b + X1 + X2)) - 3 \cdot K7 \cdot Q / (2 \cdot t^2)$$

$$= -4267 / (4 \cdot 10.8(175.0 + 52.695 + 52.695)) - 3 \cdot 0.0529 \cdot 4267 / (2 \cdot 10.8^2)$$

$$= -31.89 \text{ N./mm}^2$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$$

$$= \min(175.0 + 1.56 \cdot \sqrt{422.6 \cdot 10.8}, 2 \cdot 1260.0)$$

$$= 280.39 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	720.0000	mm.
Baseplate Thickness	Bpthk	16.0000	mm.
Baseplate Width	Bpwid	150.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	16.0000	mm.
Web Thickness	Webtk	16.0000	mm.
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$$



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$$= ((1.56 \cdot \sqrt{(417.2 \cdot 10.8)} + 225.0) \cdot 10.8^3) / 12$$

$$= 3.461 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 3.461 + 35.609 (86.831 - 5.4)^2$$

$$= 2364.725 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ² cm ⁴
Shell	329.7	10.8	5.4	35.6	19229.0	0.236E+04
Wearplate	225.0	12.0	16.8	27.0	45360.0	0.133E+04
Web	16.0	204.0	124.8	32.6	407347.2	0.160E+04
BasePlate	150.0	16.0	234.8	24.0	563519.9	0.526E+04
Totals	...	...	...	119.2	1035456.1	0.106E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 1035456 / 119.249$$

$$= 86.831 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 4266.522$$

$$= 868.3298 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 1.0181 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians})$$

$$= 660.0 - 417.2 \cdot \sin(120.0/2) / 1.0472$$

$$= 314.978 \text{ mm.}$$

Bending Moment, M = Fh * d = 273.5105 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 2.2066 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (2 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$$

$$= \sqrt{(3(4267 + 102) \cdot 150.0) / (2 \cdot 720.0 \cdot 173.33)}$$

$$= 8.789 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (414.0 + 14.0 + 12.0)$$

$$= 762.102 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nribs} - 1)$$

$$= 762.1024 / (3 - 1)$$



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= 381.051 mm.

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 381.0512 • 150.0/2
 = 285.788 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 4368.995/(720.0 • 150.0)
 = 4.045 Kg/cm²

Axial Load [P]:

= Ap • Bp
 = 285.8 • 4.05
 = 1156.119 Kg

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 25.44 + 30.484
 = 55.924 cm²

Compressive Stress [Sc]:

= P/Ar
 = 1156.1/55.9241
 = 2.027 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.111E+04
Web	190.5	16.0	8.0	30.5	24387.3	489.
Totals	...	...	...	55.9	267339.2	0.160E+04

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 175.0 - 16.0
 = 159.000 mm.

Distance to Centroid from Datum [ytot]:

= AY / A
 = 267339.25/55.924
 = 47.804 mm.

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(47.804, 175.0 - 47.804)
 = 127.196 mm.

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{1604.2/55.924}$
 = 53.558 mm.

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk
 = 660.0 - cos(120.0/2)(414.0 + 14.0 + 12.0) - 16.0
 = 424.000 mm.

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)
 = 2 • sin(120.0/2)(414.0 + 14.0 + 12.0)



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= 762.102 mm.

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance
 = 720.0 - 50.8
 = 669.200 mm.

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (762.102 - 669.2)/2
 = 46.451 mm.

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$
 = $\sqrt{(424.0^2 + 46.451^2)}$
 = 426.537 mm.

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0)}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 426.537/53.558
 = 7.964

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 60.9 / (2 • 720.0) • 381.051 • 426.54/2
 = 3.435 Kg-m.

Compressive Allowable, KL/r < Cc (7.964 < 123.2105) per AISC E2-1 [Sca]:

= (1 - (KL/r)^2 / (2 • Cc^2)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)^3 / (8 • Cc^3))
 = (1 - (7.96)^2 / (2 • 123.21^2)) 260 /
 (5/3 + 3 • (7.96) / (8 • 123.21) - (7.96^3) / (8 • 123.21^3))
 = 153.4 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 2.03/153.44 + (3.44 • 127.196/16041745) / 173.33
 = 0.015

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	159.0	95.5	25.4	242952.0	0.151E+04
Web	381.1	16.0	8.0	61.0	48774.6	418.
Totals	...	...	...	86.4	291726.5	0.192E+04

Distance to Centroid from Datum [ytot]:

= AY / A
 = 291726.531/86.408
 = 33.761 mm.

Distance to Centroid [C1]:

= max(ytot, abs(Saddle Width - ytot))
 = max(33.761, abs(175.0 - 33.761))
 = 141.239 mm.

Length of Inner Rib [L]:

= Saddle Height - Outside Radius - Bpthk
 = 660.0 - cos(440.0/2) (16.0 + 0.0 + 0.0) - 0.0



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= 204.000 mm.

Radius of Gyration [r]:
 $= \sqrt{\text{Total Inertia} / \text{Total Area}}$
 $= \sqrt{(1923.3/86.408)}$
 = 47.178 mm.

Slenderness ratio [KL/r]:
 = KL/r
 $= 1 \cdot 204.0/47.178$
 = 4.324

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 60.87 / (2 \cdot 720.0)$
 = 0.042 Kg/mm.

Moment at base of inner Rib [Mbase,c]:
 = Unit Force • e • L
 $= 0.042 \cdot 381.051 \cdot 204.0$
 = 3.286 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 = Bending Moment / Section Modulus
 $= 3.286/136171.297$
 = 0.237 N./mm²

Compressive Allowable, KL/r < Cc (4.324 < 123.2105) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot Cc^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 $= (1 - (4.32)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (4.32) / (8 \cdot 123.21) - (4.32^3) / (8 \cdot 123.21^3))$
 = 154.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$
 $= 2.89/154.68 + (3.29 \cdot 141.239/1923.262) / 173.33$
 = 0.020

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.0 mm.
User Defined Area in a Single Bolt	UserBoltArea	5.4000 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	4.2350 cm ²
Saddle Load Q0 (Weight)	Q0	3650.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	10.6 Kg
Maximum Transverse Force	Ft	261.5 Kg
Maximum Longitudinal Force	F1	60.9 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 60.9 / (4.235 \cdot 2)$
 = 0.7 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 261.5 / (4.235 \cdot 2)$
 = 3.0 N./mm². Must be less than 103.2 N./mm².



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Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= FI / (BoltShearAllowable • Nbolts)
= 60.87/(103.2 • 2.0)
= 0.0289 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= B • Ft + Sum of X Moments
= 660.0 • 261.46 + 0.0
= 172.57 Kg-m.

Eccentricity (e):
= Rmom / QO
= 172.57/3649.98
= 47.28 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 0.0289, 0.0]
= 0.0289 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: N2- Channel outlet

From: 20

Pressure for Reinforcement Calculations	P	5.516	bars
Temperature for Internal Pressure	Temp	71	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	828.00	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	3.2000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		350.00	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

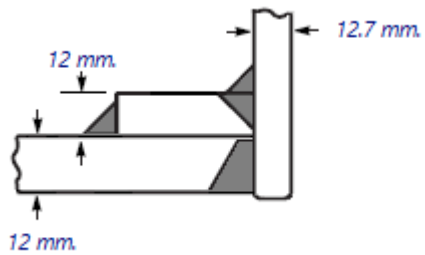
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.2000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	319.0750	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	10.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		50.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N2- Channel outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	8.625 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.52 \cdot 417.2 / (138 \cdot 1.0 - 0.6 \cdot 5.52)$
 $= 1.6717 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5.52 \cdot 109.5375 / (118 \cdot 1.0 + 0.4 \cdot 5.52)$
 $= 0.5111 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	400.1500 mm.
Parallel to Vessel Wall, opening length	d	200.0750 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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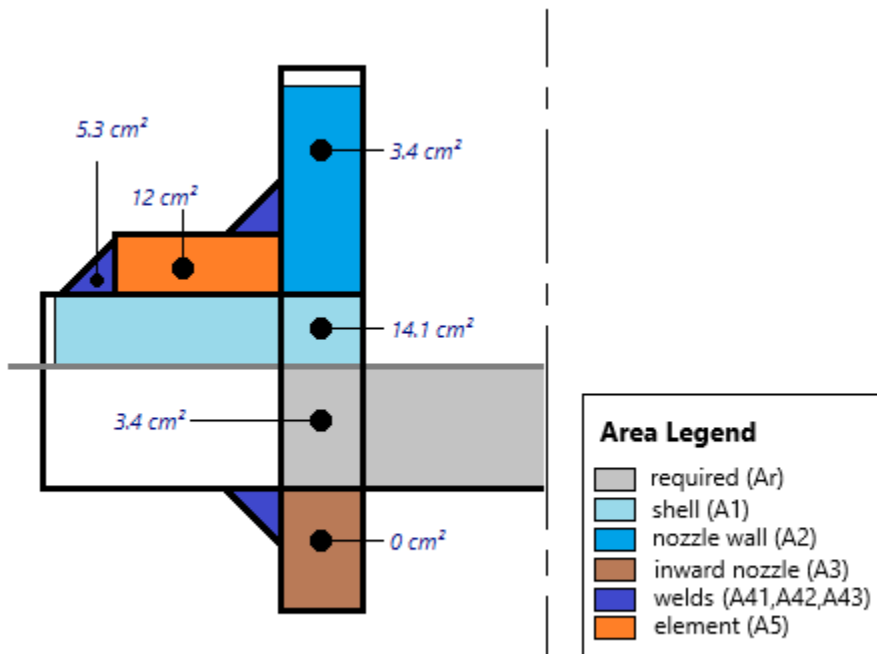


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3.391	NA	NA
Area in Shell	A1	14.066	NA	NA
Area in Nozzle Wall	A2	3.382	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	5.275	NA	NA
Area in Element	A5	12.000	NA	NA
TOTAL AREA AVAILABLE	Atot	34.723	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (200.075 \cdot 1.6717 \cdot 1.0 + 2 \cdot 9.5 \cdot 1.6717 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 3.391 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 200.075(1.0 \cdot 8.8 - 1.0 \cdot 1.672) - 2 \cdot 9.5$$

$$(1.0 \cdot 8.8 - 1.0 \cdot 1.6717) \cdot (1 - 0.855)$$

$$= 14.066 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 22.0) \cdot (9.5 - 0.51) \cdot 0.8551$$



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= 3.382 cm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Wp²•fr4
 = (5.0) • 0.86 + (0.0) • 0.86 + 10.0² • 1.0
 = 5.275 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (319.075 - 219.075) • 12.0 • 1.0
 = 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.7111 mm.
 Wall Thickness per UG16(b), tr16b = 5.7000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 4.8717 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 5.7000 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 5.7000 mm.
 Wall Thickness per table UG-45 tb3 = 10.3600 mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[10.36, max(5.7, 5.7)]
 = 5.7000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.7111, 5.7)
 = 5.7000 mm.

Available Nozzle Neck Thickness = 0.875 * 12.700 = 11.113 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	27.4,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	267.6 N./mm ²	Passed
Occasional	:	2.8,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	17.2,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.113, tr = 0.511, c = 3.2 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.065, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11.113, tr = 0.511, c = 3.2 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.065, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Govrn. thk, tg = 11.113, tr = 0.511, c = 3.2 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.065$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, tr = 1.672, c = 3.2 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.19$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 11.113, tr = 0.511, c = 3.2 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.065$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5.52 / 19.60 = 0.281$

Weld Size Calculations, Description: N2- Channel out

Intermediate Calc. for nozzle/shell Welds	Tmin	9.5000 mm.
Intermediate Calc. for pad/shell Welds	TminPad	8.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$21.2100 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$4.4000 = 0.5 \cdot T_{minPad}$	$7.0700 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (3.3907 - 14.0656 + 2 \cdot 9.5 \cdot 0.8551 \cdot (1.0 \cdot 8.8 - 1.6717)) \cdot 138)$
 $= \max(0, -13391.60) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / 707)^2 \cdot f_r2) \cdot S_v$
 $= (3.3819 + 12.0 + 5.2754 - 0.0 \cdot 0.86) \cdot 138$
 $= 29068.02 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (3.3819 + 0.0 + 7.1484 + (1.4297)) \cdot 138$



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= 16829.58 Kgf

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S$$

$$= (3.3819 + 0.0 + 5.2754 + 12.0 + (1.4297)) \cdot 138$$

$$= 31079.80 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 219.075 \cdot 30.0 \cdot 0.49 \cdot 118$$

$$= 60866. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 319.075 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 34558. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 104.7875) \cdot (12.7 - 3.2) \cdot 0.7 \cdot 118$$

$$= 26341. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 219.075 \cdot 12.0 \cdot 0.74 \cdot 138$$

$$= 43000. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (12.0 - 3.2) \cdot 0.74 \cdot 138$$

$$= 31533. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (34558 + 26341) = 60899 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (60866 + 43000 + 31533 + 0) = 135399 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (34558 + 31533 + 0) = 66092 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 60898 Kgf, must exceed W = 0 Kgf	or W1 = 29068 Kgf
Path 2-2 = 135399 Kgf, must exceed W = 0 Kgf	or W2 = 16829 Kgf
Path 3-3 = 66091 Kgf, must exceed W = 0 Kgf	or W3 = 31079 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 29 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 14.7538 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 276.7538 mm.

This nozzle or its diameter limit intersects weld seam # [3] on this element. The distance between the nozzle and seam is, 116.50 mm., while the distance from the center of the nozzle to the edge of the diameter limit is: 200.07 mm.



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Input Echo, WRC297 Item 1, Description: N2- Channel ou:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	828.000	mm.
Shell Thickness	12.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.2000	mm.
Nozzle Diameter	219.075	mm.
Nozzle Thickness	12.7000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	319.075	mm.
Thickness of Reinforcing Pad	12.0000	mm.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	5.52	bars
Radial Load	P	-550.68	Kgf
Circumferential Shear	Vc	815.83	Kgf
Longitudinal Shear	Vl	815.83	Kgf
Circumferential Moment	Mc	489.31	Kg-m.
Longitudinal Moment	Ml	652.41	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{427.6(12.0 - 3.2)}$
 $= 421.505 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	855.200	mm.
Nozzle Outside Diameter	(d) =	219.075	mm.
Vessel Thickness used	(T) =	20.800	mm.
Nozzle Thickness used	(t) =	9.500	mm.
T / t	=	2.189	
d / t	=	23.061	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.643$$



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Nr/P = 0.080
 Mr/P = 0.088
 MØ/P = 0.057
 NØ/P = 0.161
 MrD/Mc = 0.192
 NrDL/Mc = 0.128
 MØd/Mc = 0.145
 NØDL/Mc = 0.154
 MrD/Ml = 0.130
 NrDL/Ml = 0.085
 MØD/Ml = 0.094
 NØDL/Ml = 0.273

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-38	38	38	-38
Normal Pressure Stress	10	11	10	11
Outplane Stress Summary	-39	28	73	-10

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	5	5	5	5
Inplane Stress Summary	-45	46	71	-47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	49	48	76	47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	D1 Right
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circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	5	5	5	5
Outplane Stress Summary	-40	39	63	-33

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-58	58	58	-58
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	10	11	10	11
Inplane Stress Summary	-46	57	83	-46

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	48	58	85	48

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	-9	-9	27	27

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-170	170	170	-170
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-168	136	211	-162

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11
Shear Stress Summary	13	13	8	8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	170	148	212	191

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	0	1	16	17

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-187	187	187	-187
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-180	158	223	-184

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	11	11	11	11
Shear Stress Summary	8	8	13	13



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	182	159	224	203

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(421.6(12.0 - 3.2))}$
 $= 420.079 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	843.200 mm.
Nozzle Outside Diameter	(d) =	319.075 mm.
Vessel Thickness used	(T) =	8.800 mm.
Nozzle Thickness used	(t) =	59.500 mm.
T / t	=	0.148
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.704$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.053
Mr/P	=	0.089
M0/P	=	0.026
N0/P	=	0.045
MrD/Mc	=	0.257
NrDL/Mc	=	0.098
M0d/Mc	=	0.077
N0DL/Mc	=	0.073
MrD/Ml	=	0.068
NrDL/Ml	=	0.059
M0D/Ml	=	0.020
N0DL/Ml	=	0.051

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	25	26	25	26
Outplane Stress Summary	-4	37	84	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



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Inplane Membrane (ML)	-15	-15	15	15
Inplane Bending (ML)	-105	105	105	-105
Inplane Pressure Stress	12	12	12	12
Inplane Stress Summary	-67	70	174	-111

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	7	7	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	68	71	175	111

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-89	89	89	-89
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	12	12	12	12
Outplane Stress Summary	-76	80	130	-70

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	-19	-19	19	19
Inplane Bending (Mc)	-299	299	299	-299
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	25	26	25	26
Inplane Stress Summary	-252	273	385	-287

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	3	3	7	7



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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	252	273	385	288

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	11	10	11	10	11	10	11
Circ. Pl (SUS)	-17	-17	19	19	-6	-6	6	6
Circ. Q (SUS)	-34	34	42	-42	-52	52	64	-64
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-5	-5	5	5	-6	-6	8	8
Long. Q (SUS)	-46	46	58	-58	-40	40	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	10.0	11.0	10.0	11.0	10.0	11.0	10.0	11.0
Pm+Pl (SUS)	7.3	6.3	29.1	30.0	5.4	6.3	16.3	17.2
Pm+Pl+Q (Total)	50.0	47.8	73.8	48.4	49.8	57.8	81.7	49.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	138.00	Passed
Pm+Pl (SUS)	30.05	207.00	Passed
Pm+Pl+Q (TOTAL)	81.74	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	25	26	25	26	25	26	25	26
Circ. Pl (SUS)	-10	-10	16	16	-16	-16	22	22
Circ. Q (SUS)	-20	20	42	-42	-262	262	336	-336
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-12	-12	18	18	-11	-11	17	17
Long. Q (SUS)	-68	68	142	-142	-78	78	100	-100
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	25.0	26.0	25.0	26.0	25.0	26.0	25.0	26.0



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Pm+Pl (SUS)	15.1	16.1	41.1	42.1	9.1	10.1	47.1	48.1
Pm+Pl+Q (Total)	68.6	69.1	172.2	112.3	253.1	272.1	383.1	288.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26.00	138.00	Passed
Pm+Pl (SUS)	48.05	207.00	Passed
Pm+Pl+Q (TOTAL)	383.14	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. Pl (SUS)	-17	-17	19	19	-6	-6	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-20	-20	20	20	-15	-15	15	15
Long. Q (SUS)	-153	153	187	-187	-170	170	204	-204
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+Pl (SUS)	17.6	17.5	26.2	27.0	12.6	13.6	18.8	19.0
Pm+Pl+Q (Total)	171.1	148.3	210.4	190.9	182.9	158.5	222.8	202.7

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+Pl (SUS)	27.00	177.00	Passed
Pm+Pl+Q (TOTAL)	222.81	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: N1- Channel inlet

From: 20

Pressure for Reinforcement Calculations	P	5.594	bars
Temperature for Internal Pressure	Temp	71	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	828.00	mm.
Shell Finished (Minimum) Thickness	t	12.0000	mm.
Shell Internal Corrosion Allowance	c	3.2000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		350.00	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

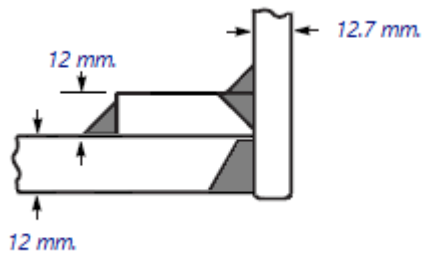
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.2000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	30.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	12.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	319.0750	mm.
Thickness of Pad	te	12.0000	mm.
Weld leg size between Pad and Shell	Wp	10.0000	mm.
Groove weld depth between Pad and Nozzle	Wgpn	12.0000	mm.
Reinforcing Pad Width		50.0000	mm.
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: N1- Channel inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	8.625 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 5.59 \cdot 417.2 / (138 \cdot 1.0 - 0.6 \cdot 5.59)$
 $= 1.6956 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 5.59 \cdot 109.5375 / (118 \cdot 1.0 + 0.4 \cdot 5.59)$
 $= 0.5184 \text{ mm.}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	400.1500 mm.
Parallel to Vessel Wall, opening length	d	200.0750 mm.
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		22.0000 mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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Nozzle Calcs.: N1- Channel in1 Nozl: 10 9:55am Aug 12,2026

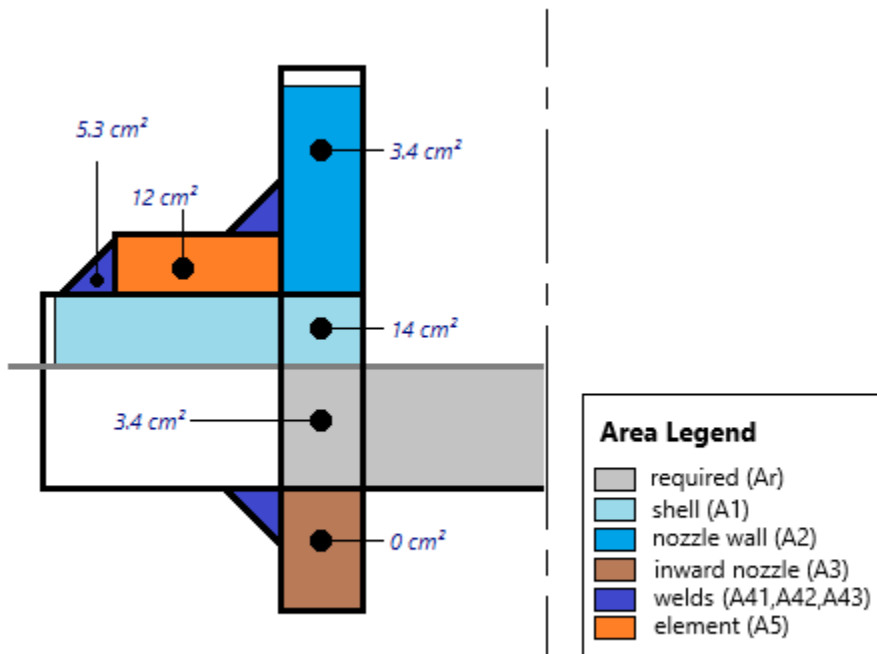


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3.439	NA	NA
Area in Shell	A1	14.019	NA	NA
Area in Nozzle Wall	A2	3.379	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	5.275	NA	NA
Area in Element	A5	12.000	NA	NA
TOTAL AREA AVAILABLE	Atot	34.673	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (200.075 \cdot 1.6956 \cdot 1.0 + 2 \cdot 9.5 \cdot 1.6956 \cdot 1.0 \cdot (1 - 0.86)) \\
 &= 3.439 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 200.075(1.0 \cdot 8.8 - 1.0 \cdot 1.696) - 2 \cdot 9.5 \\
 &\quad (1.0 \cdot 8.8 - 1.0 \cdot 1.6956) \cdot (1 - 0.855) \\
 &= 14.019 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 22.0) \cdot (9.5 - 0.52) \cdot 0.8551
 \end{aligned}$$



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= 3.379 cm²

Area Available in Welds [A41 + A42 + A43]:
 = (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Wp²•fr4
 = (5.0) • 0.86 + (0.0) • 0.86 + 10.0² • 1.0
 = 5.275 cm²

Area Available in Element [A5]:
 = (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (319.075 - 219.075) • 12.0 • 1.0
 = 12.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.7184 mm.
 Wall Thickness per UG16(b), tr16b = 5.7000 mm.
 Wall Thickness, shell/head, internal pressure trb1 = 4.8956 mm.
 Wall Thickness tb1 = max(trb1, tr16b) = 5.7000 mm.
 Wall Thickness tb2 = max(trb2, tr16b) = 5.7000 mm.
 Wall Thickness per table UG-45 tb3 = 10.3600 mm.

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1,tb2)]
 = min[10.36, max(5.7, 5.7)]
 = 5.7000 mm.

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(3.7184, 5.7)
 = 5.7000 mm.

Available Nozzle Neck Thickness = 0.875 * 12.700 = 11.113 mm. -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	27.4,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	267.6 N./mm ²	Passed
Occasional	:	2.8,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	17.2,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 11.113, tr = 0.518, c = 3.2 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.066, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-5 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 11.113, tr = 0.518, c = 3.2 mm., E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.066, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Govrn. thk, tg = 11.113, tr = 0.518, c = 3.2 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.066$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 12.0, tr = 1.696, c = 3.2 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.193$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 11.113, tr = 0.518, c = 3.2 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.066$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $5.59/19.60 = 0.285$

Weld Size Calculations, Description: N1- Channel inl

Intermediate Calc. for nozzle/shell Welds	Tmin	9.5000 mm.
Intermediate Calc. for pad/shell Welds	TminPad	8.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$21.2100 = 0.7 \cdot W_o \text{ mm.}$
Pad Weld	$4.4000 = 0.5 \cdot T_{minPad}$	$7.0700 = 0.7 \cdot W_p \text{ mm.}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (3.4391 - 14.0186 + 2 \cdot 9.5 \cdot 0.8551 \cdot (1.0 \cdot 8.8 - 1.6956)) \cdot 138)$
 $= \max(0, -13262.90) \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / 707)^2 \cdot f_r2) \cdot S_v$
 $= (3.3792 + 12.0 + 5.2754 - 0.0 \cdot 0.86) \cdot 138$
 $= 29064.18 \text{ Kg}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (3.3792 + 0.0 + 7.1484 + (1.4297)) \cdot 138$



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= 16825.74 Kgf

Weld Load [W3]:

= (A2+A3+A4+A5+(2*tn*t*fr1))*S
 = (3.3792 + 0.0 + 5.2754 + 12.0 + (1.4297)) • 138
 = 31075.96 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

= (π/2) • Dlo • Wo • 0.49 • Snw
 = (3.1416/2.0) • 219.075 • 30.0 • 0.49 • 118
 = 60866. Kgf

Shear, Pad Element Weld [Spew]:

= (π/2) • DP • WP • 0.49 • SEW
 = (3.1416/2.0) • 319.075 • 10.0 • 0.49 • 138
 = 34558. Kgf

Shear, Nozzle Wall [Snw]:

= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
 = (3.1416 • 104.7875) • (12.7 - 3.2) • 0.7 • 118
 = 26341. Kgf

Tension, Pad Groove Weld [Tpgw]:

= (π/2) • Dlo • Wgpn • 0.74 • Seg
 = (3.1416/2) • 219.075 • 12.0 • 0.74 • 138
 = 43000. Kgf

Tension, Shell Groove Weld [Tngw]:

= (π/2) • Dlo • (T - cas) • 0.74 • Sng
 = (3.1416/2.0) • 219.075 • (12.0 - 3.2) • 0.74 • 138
 = 31533. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SPEW + SNW) = (34558 + 26341) = 60899 Kgf

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)
 = (60866 + 43000 + 31533 + 0) = 135399 Kgf

Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)
 = (34558 + 31533 + 0) = 66092 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 60898 Kgf, must exceed W = 0 Kgf	or W1 = 29064 Kgf
Path 2-2 = 135399 Kgf, must exceed W = 0 Kgf	or W2 = 16825 Kgf
Path 3-3 = 66091 Kgf, must exceed W = 0 Kgf	or W3 = 31075 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 29 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 14.7538 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 276.7538 mm.

Input Echo, WRC297 Item 1, Description: N1- Channel in:

Diameter Basis for Cylindrical Shell

ID



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Shell Internal Corrosion Allowance		3.2000	mm.
Shell External Corrosion Allowance		0.0000	mm.
Shell Diameter		828.000	mm.
Shell Thickness		12.0000	mm.
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-516 70	
Vessel Cold S.I. Allowable	Smc	138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle		OD	
Nozzle Corrosion Allowance		3.2000	mm.
Nozzle Diameter		219.075	mm.
Nozzle Thickness		12.7000	mm.
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad		319.075	mm.
Thickness of Reinforcing Pad		12.0000	mm.

Note:
 External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	5.59	bars
Radial Load	P	-550.68	Kgf
Circumferential Shear	Vc	815.83	Kgf
Longitudinal Shear	Vl	815.83	Kgf
Circumferential Moment	Mc	489.31	Kg-m.
Longitudinal Moment	Ml	652.41	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{427.6(12.0 - 3.2)}$$

$$= 421.505 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	855.200	mm.
Nozzle Outside Diameter	(d) =	219.075	mm.
Vessel Thickness used	(T) =	20.800	mm.
Nozzle Thickness used	(t) =	9.500	mm.
T / t	=	2.189	
d / T	=	23.061	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.643$$

Nr/P	=	0.080
Mr/P	=	0.088
M0/P	=	0.057



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NØ/P = 0.161
 MrD/Mc = 0.192
 NrDL/Mc = 0.128
 MØd/Mc = 0.145
 NØDL/Mc = 0.154
 MrD/Ml = 0.130
 NrDL/Ml = 0.085
 MØD/Ml = 0.094
 NØDL/Ml = 0.273

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-38	38	38	-38
Normal Pressure Stress	10	11	10	11
Outplane Stress Summary	-39	28	73	-10

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-52	52	52	-52
Inplane Pressure Stress	5	5	5	5
Inplane Stress Summary	-45	46	71	-47

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5
Shear Stress Summary	6	6	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	49	48	76	47

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4



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Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	5	5	5	5
Outplane Stress Summary	-40	39	63	-33

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-58	58	58	-58
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	10	11	10	11
Inplane Stress Summary	-46	57	83	-46

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5
Shear Stress Summary	3	3	6	6

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	48	58	85	48

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7
Hoop Stress Summary	-9	-9	27	27

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0



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 Nozzle Calcs.: N1- Channel inl Nozl: 10 9:55am Aug 12,2026

Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-170	170	170	-170
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-168	136	211	-162
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11

Shear Stress Summary	13	13	8	8
----------------------	----	----	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	170	148	212	191
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	6	7	6	7

Hoop Stress Summary	0	1	16	17
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	17	-17	17	-17
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-187	187	187	-187
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3

Axial Stress Summary	-180	158	223	-184
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	11	11	11	11

Shear Stress Summary	8	8	13	13
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside



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 Nozzle Calcs.: N1- Channel inl Nozl: 10 9:55am Aug 12,2026

Two * Max Shear Stress	182	159	224	203
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{421.6(12.0 - 3.2)}$
 $= 420.079 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	843.200 mm.
Nozzle Outside Diameter	(d) =	319.075 mm.
Vessel Thickness used	(T) =	8.800 mm.
Nozzle Thickness used	(t) =	59.500 mm.
T / t	=	0.148
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.704$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.053
Mr/P	=	0.089
M0/P	=	0.026
N0/P	=	0.045
MrD/Mc	=	0.257
NrDL/Mc	=	0.098
M0d/Mc	=	0.077
N0DL/Mc	=	0.073
MrD/Ml	=	0.068
NrDL/Ml	=	0.059
M0D/Ml	=	0.020
N0DL/Ml	=	0.051

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-31	31	31	-31
Normal Pressure Stress	26	26	26	26
Outplane Stress Summary	-4	37	85	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-15	-15	15	15
Inplane Bending (ML)	-105	105	105	-105
Inplane Pressure Stress	13	13	13	13



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 Nozzle Calcs.: N1- Channel inl Nozl: 10 9:55am Aug 12,2026

Inplane Stress Summary -67 70 175 -111

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE Au Al Bu Bl
 (Shear stress normal to Top Top Bottom Bottom
 longitudinal plane) Outside Inside Outside Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	5	5	5	5

Shear Stress Summary 7 7 3 3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE Au Al Bu Bl
 (Stress Intensities Top Top Bottom Bottom
 (Stress Intensities Outside Inside Outside Inside

Two * Max Shear Stress 68 71 175 111

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stresses Normal to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-14	-14	14	14
Outplane Bending (Mc)	-89	89	89	-89
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	13	13	13	13

Outplane Stress Summary -76 80 131 -70

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stresses parallel to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	37	-37	37	-37
Inplane Membrane (Mc)	-19	-19	19	19
Inplane Bending (Mc)	-299	299	299	-299
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	26	26	26	26

Inplane Stress Summary -251 273 385 -287

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Shear stress normal to Left Left Right Right
 circumferential plane) Outside Inside Outside Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	5	5	5	5

Shear Stress Summary 3 3 7 7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE Cu Cl Du Dl
 (Stress Intensities Left Left Right Right
 (Stress Intensities Outside Inside Outside Inside



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 Nozzle Calcs.: N1- Channel inl Nozl: 10 9:55am Aug 12,2026

Two * Max Shear Stress 251 274 386 287

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	11	10	11	10	11	10	11
Circ. Pl (SUS)	-17	-17	19	19	-6	-6	6	6
Circ. Q (SUS)	-34	34	42	-42	-52	52	64	-64
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-5	-5	5	5	-6	-6	8	8
Long. Q (SUS)	-46	46	58	-58	-40	40	48	-48
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	10.0	11.0	10.0	11.0	10.0	11.0	10.0	11.0
Pm+Pl (SUS)	7.3	6.3	29.1	30.0	5.4	6.3	16.3	17.2
Pm+Pl+Q (Total)	50.0	47.8	73.8	48.4	49.8	57.8	81.7	49.5

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	138.00	Passed
Pm+Pl (SUS)	30.05	207.00	Passed
Pm+Pl+Q (TOTAL)	81.74	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	26	26	26	26	26	26	26	26
Circ. Pl (SUS)	-10	-10	16	16	-16	-16	22	22
Circ. Q (SUS)	-20	20	42	-42	-262	262	336	-336
Long. Pm (SUS)	13	13	13	13	13	13	13	13
Long. Pl (SUS)	-12	-12	18	18	-11	-11	17	17
Long. Q (SUS)	-68	68	142	-142	-78	78	100	-100
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	5	5	5	5	5	5	5	5
Pm (SUS)	26.0	26.0	26.0	26.0	26.0	26.0	26.0	26.0
Pm+Pl (SUS)	16.1	16.1	42.1	42.1	10.1	10.1	48.1	48.1
Pm+Pl+Q (Total)	67.6	70.1	173.2	111.3	252.1	272.1	384.1	288.2



Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26.00	138.00	Passed
Pm+Pl (SUS)	48.06	207.00	Passed
Pm+Pl+Q (TOTAL)	384.14	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	6	7	6	7	6	7	6	7
Circ. Pl (SUS)	-17	-17	19	19	-6	-6	8	8
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-20	-20	20	20	-15	-15	15	15
Long. Q (SUS)	-153	153	187	-187	-170	170	204	-204
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	6.0	7.0	6.0	7.0	6.0	7.0	6.0	7.0
Pm+Pl (SUS)	17.6	17.5	26.2	27.0	12.6	13.6	18.8	19.0
Pm+Pl+Q (Total)	171.1	148.3	210.4	190.9	182.9	158.5	222.8	202.7

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+Pl (SUS)	27.00	177.00	Passed
Pm+Pl+Q (TOTAL)	222.81	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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 Nozzle Calcs.: Shell inlet Nozl: 11 9:55am Aug 12,2026

Input, Nozzle Desc: Shell inlet

From: 50

Pressure for Reinforcement Calculations	P	12.653	bars
Temperature for Internal Pressure	Temp	143	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	828.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	c	3.2000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1533.50	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

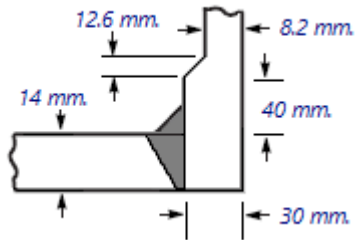
Type of Element Connected to the Parent : Nozzle

Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		8.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	40.0000	mm.
Height of Beveled Transition	L'	12.5985	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		30.0000	mm.
Corrosion Allowance	can	3.2000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		6.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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 Nozzle Calcs.: Shell inlet Nozl: 11 9:55am Aug 12,2026



Reinforcement CALCULATION, Description: Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 8.625 in.
 Actual Thickness Used in Calculation 0.322 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.65 \cdot 417.2 / (138 \cdot 1.0 - 0.6 \cdot 12.65)$
 $= 3.8467 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 12.65 \cdot 109.5375 / (138 \cdot 1.0 + 0.4 \cdot 12.65)$
 $= 1.0007 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 21.821 / 0.5773$
 $= 37.7954 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	418.2348	mm.
Parallel to Vessel Wall, opening length	d	209.1174	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.0000	mm.

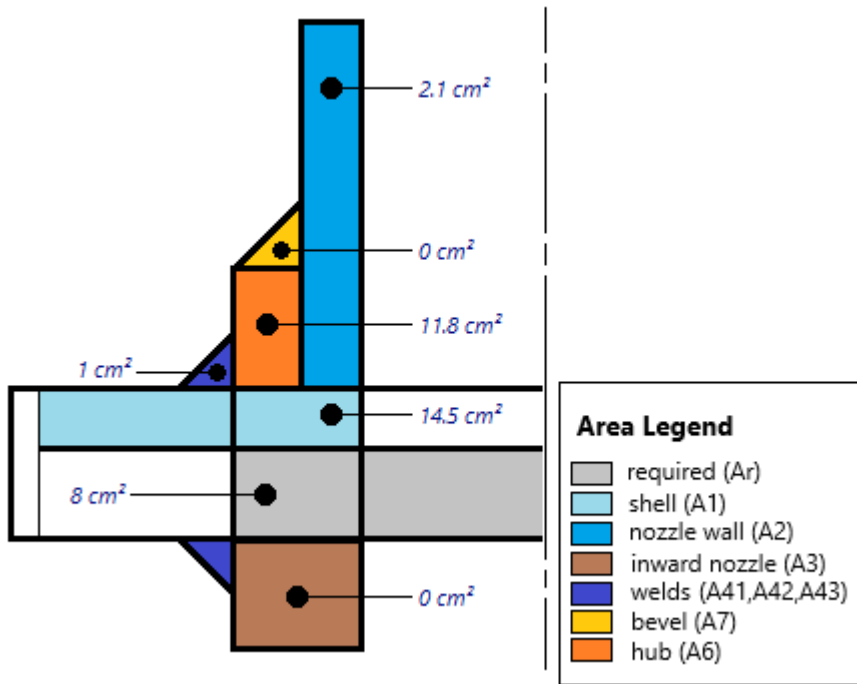
Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	8.044	NA	NA
Area in Shell	A1	14.541	NA	NA
Area in Nozzle Wall	A2	2.148	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	11.783	NA	NA
TOTAL AREA AVAILABLE	Atot	29.472	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (209.1174 \cdot 3.8467 \cdot 1.0 + 2 \cdot 26.8 \cdot 3.8467 \cdot 1.0 \cdot (1 - 1.0))$
 $= 8.044 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 209.117(1.0 \cdot 10.8 - 1.0 \cdot 3.847) - 2 \cdot 26.8$
 $(1.0 \cdot 10.8 - 1.0 \cdot 3.8467) \cdot (1 - 1.0)$
 $= 14.541 \text{ cm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 27.0)(4.98 - 1.0)1.0$
 $= 2.148 \text{ cm}^2$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(27.0, 250.0, 40.0)) \cdot (30.0 - 8.2) \cdot 1.0$$

$$= 11.783 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 4.2007 mm.
Wall Thickness per UG16(b),	tr16b = 4.7000 mm.
Wall Thickness, shell/head, internal pressure	trb1 = 7.0467 mm.
Wall Thickness	tb1 = max(trb1, tr16b) = 7.0467 mm.
Wall Thickness	tb2 = max(trb2, tr16b) = 4.7000 mm.
Wall Thickness per table UG-45	tb3 = 10.3600 mm.

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb3, \max(tb1, tb2)]$$

$$= \min[10.36, \max(7.0467, 4.7)]$$

$$= 7.0467 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(ta, tb)$$

$$= \max(4.2007, 7.0467)$$

$$= 7.0467 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 8.179 = 7.156 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 57.0,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 288.0 N./mm ²	Passed
Occasional	: 13.0,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 30.3,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.156, tr = 1.001, c = 3.2 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.253$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 3.847, c = 3.2 mm., E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.356$, Temp. Reduction = 73 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:



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Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 12.65/51.00 = 0.248

Weld Size Calculations, Description: Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 10.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (8.0441 - 14.5406 + 2 \cdot 26.8 \cdot 1.0 \cdot (1.0 \cdot 10.8 - 3.8467)) \cdot 138)$
 $= \max(0, -3897.11) \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (13.9316 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 21011.11 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (13.9316 + 0.0 + 1. + (5.7888)) \cdot 138$
 $= 29156.86 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (13.9316 + 0.0 + 1. + 0.0 + (5.7888)) \cdot 138$
 $= 29156.86 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 23727. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 107.0481) \cdot (30.0 - 3.2) \cdot 0.7 \cdot 138$
 $= 88778. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 219.075 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 21500. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (23727 + 88778) = 112505 \text{ Kgf}$



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Failure Path [Path 2-2]:
 = (Sonw + Tpgw + Tngw + Sinw)
 = (23727 + 0 + 21500 + 0) = 45227 Kgf

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (23727 + 21500 + 0) = 45227 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 112505 Kgf, must exceed W = 0 Kgf	or W1 = 21011 Kgf
Path 2-2 = 45227 Kgf, must exceed W = 0 Kgf	or W2 = 29156 Kgf
Path 3-3 = 45227 Kgf, must exceed W = 0 Kgf	or W3 = 29156 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 13 bars

The Drop for this Nozzle is : 14.7538 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 278.7538 mm.

Input Echo, WRC297 Item 1, Description: Shell inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	828.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.2000	mm.
Nozzle Diameter	262.717	mm.
Nozzle Thickness	30.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 4	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	12.65	bars
Radial Load	P	-550.68	Kgf
Circumferential Shear	Vc	815.83	Kgf
Longitudinal Shear	Vl	815.83	Kgf
Circumferential Moment	Mc	489.31	Kg-m.
Longitudinal Moment	Ml	652.41	Kg-m.
Torsional Moment	Mt	815.51	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (8.757) is < 10
 check the Limitations of WRC bulletin 297.



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Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 262.717 + 2 \cdot 1.65 \cdot \sqrt{422.6(14.0 - 3.2)}$
 $= 485.659 \text{ mm.}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 845.200 mm.
 Nozzle Outside Diameter (d) = 262.717 mm.
 Vessel Thickness used (T) = 10.800 mm.
 Nozzle Thickness used (t) = 26.800 mm.
 T / t = 0.403
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.750$$

Nr/P = 0.061
 Mr/P = 0.112
 M0/P = 0.035
 N0/P = 0.080
 MrD/Mc = 0.269
 NrDL/Mc = 0.098
 M0d/Mc = 0.081
 N0DL/Mc = 0.084
 MrD/Ml = 0.095
 NrDL/Ml = 0.057
 M0D/Ml = 0.027
 N0DL/Ml = 0.094

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-19	-19	19	19
Outplane Bending (ML)	-33	33	33	-33
Normal Pressure Stress	48	49	48	49
Outplane Stress Summary	8	57	114	29

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-119	119	119	-119
Inplane Pressure Stress	24	24	24	24
Inplane Stress Summary	-72	103	188	-111

Vessel Stresses (N/mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6

Shear Stress Summary	8	8	5	5
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	83	104	189	141
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-76	76	76	-76
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	24	24	24	24

Outplane Stress Summary	-51	81	126	-45
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	-15	-15	15	15
Inplane Bending (Mc)	-252	252	252	-252
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	48	49	48	49

Inplane Stress Summary	-185	258	350	-216
------------------------	------	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	5	5	8	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	186	258	350	216
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom



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hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-19	-19	19	19
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	-10	-9	28	30

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-5	-5	5	5
Axial Bending (ML)	-13	13	13	-13
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-13	6	26	-8

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	3	3	2	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	15	17	29	38

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	5	6	5	6
Hoop Stress Summary	-4	-2	22	23

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	3	-3	3	-3



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Axial Membrane (Mc)	-4	-4	4	4
Axial Bending (Mc)	-33	33	33	-33
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-31	28	44	-29

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	2	2	2	2
Shear Stress Summary	2	2	3	3

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	31	31	45	53

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	48	49	48	49	48	49	48	49
Circ. Pl (SUS)	-16	-16	22	22	-13	-13	17	17
Circ. Q (SUS)	-24	24	42	-42	-221	221	283	-283
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	-9	-9	13	13	-10	-10	16	16
Long. Q (SUS)	-88	88	150	-150	-67	67	85	-85
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	48.0	49.0	48.0	49.0	48.0	49.0	48.0	49.0
Pm+Pl (SUS)	32.1	33.1	70.0	71.0	35.0	36.0	65.0	66.0
Pm+Pl+Q (Total)	82.2	104.0	187.3	142.4	186.2	257.1	348.2	217.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	49.00	138.00	Passed
Pm+Pl (SUS)	71.03	207.00	Passed
Pm+Pl+Q (TOTAL)	348.22	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):



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Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	5	6	5	6	5	6	5	6
Circ. Pl (SUS)	-16	-16	22	22	-10	-10	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-5	-5	5	5	-4	-4	4	4
Long. Q (SUS)	-10	10	16	-16	-30	30	36	-36
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	5.0	6.0	5.0	6.0	5.0	6.0	5.0	6.0
Pm+Pl (SUS)	11.0	10.0	27.0	28.0	5.0	4.0	21.0	22.0
Pm+Pl+Q (Total)	14.2	17.5	27.8	37.2	32.1	32.2	42.2	52.2

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	6.00	138.00	Passed
Pm+Pl (SUS)	28.00	207.00	Passed
Pm+Pl+Q (TOTAL)	52.15	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: Shell outlet

From: 50

Pressure for Reinforcement Calculations	P	12.653	bars
Temperature for Internal Pressure	Temp	143	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	828.00	mm.
Shell Finished (Minimum) Thickness	t	14.0000	mm.
Shell Internal Corrosion Allowance	c	3.2000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		6933.50	mm.
User Entered Minimum Design Metal Temperature		11.40	°C

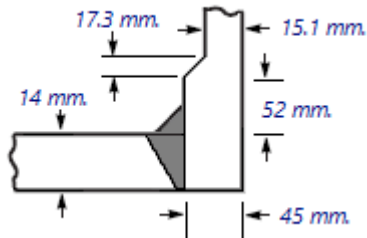
Type of Element Connected to the Parent : Nozzle

Material		SA-266 4	
Material UNS Number		K03017	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		10.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	52.0000	mm.
Height of Beveled Transition	L'	17.2699	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		45.0000	mm.
Corrosion Allowance	can	3.2000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	250.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		6.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	10.750 in.
Actual Thickness Used in Calculation	0.594 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 12.65 \cdot 417.2 / (138 \cdot 1.0 - 0.6 \cdot 12.65)$
 $= 3.8468 \text{ mm.}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 12.65 \cdot 136.525 / (138 \cdot 1.0 + 0.4 \cdot 12.65)$
 $= 1.2473 \text{ mm.}$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 29.912 / 0.5773$
 $= 51.8098 \text{ mm.}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	498.5496	mm.
Parallel to Vessel Wall, opening length	d	249.2748	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	27.0000	mm.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

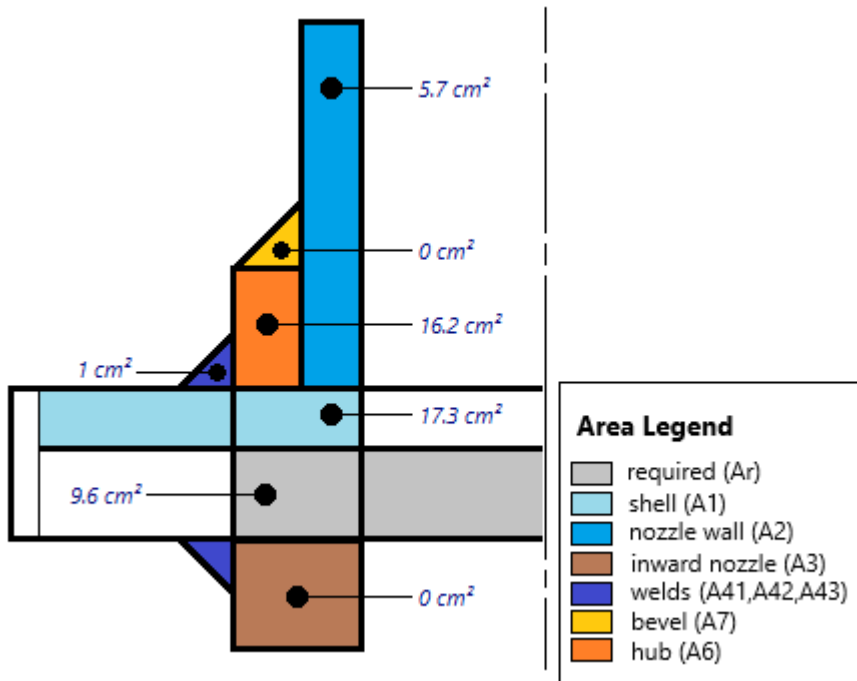


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Nozzle Calcs.: Shell outlet Nozl: 12 9:55am Aug 12,2026



Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	9.589	NA	NA
Area in Shell	A1	17.333	NA	NA
Area in Nozzle Wall	A2	5.746	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	16.153	NA	NA
TOTAL AREA AVAILABLE	Atot	40.231	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (249.2748 \cdot 3.8468 \cdot 1.0 + 2 \cdot 41.8 \cdot 3.8468 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 9.589 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 249.275(1.0 \cdot 10.8 - 1.0 \cdot 3.847) - 2 \cdot 41.8$$

$$(1.0 \cdot 10.8 - 1.0 \cdot 3.8468) \cdot (1 - 1.0)$$

$$= 17.333 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot tlnp)(tn - trn)fr2$$

$$= (2 \cdot 27.0)(11.89 - 1.25)1.0$$

$$= 5.746 \text{ cm}^2$$



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Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot fr_2 + (W_i - can / 0.707)^2 \cdot fr_2$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(T_{lnp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot fr_2$$

$$= (2 \cdot \min(27.0, 250.0, 52.0)) \cdot (45.0 - 15.1) \cdot 1.0$$

$$= 16.153 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 4.4473 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.7000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 7.0468 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 7.0468 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.7000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 11.3026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.303, \max(7.0468, 4.7)]$$

$$= 7.0468 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.4473, 7.0468)$$

$$= 7.0468 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 15.088 = 13.202 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 26.1,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 318.9 N./mm ²	Passed
Occasional	: 6.3,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 13.8,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 13.202$, $tr = 1.247$, $c = 3.2 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.125$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A	0°C
Min Metal Temp. at Required thickness (UCS 66.1)	-104°C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: B

Govrn. thk, $t_g = 14.0$, $tr = 3.847$, $c = 3.2 \text{ mm.}$, $E^* = 1.0$
 Thickness Ratio = $tr \cdot E^* / (t_g - c) = 0.356$, Temp. Reduction = 73°C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18°C
Min Metal Temp. at Required thickness (UCS 66.1)	-48°C
Min Metal Temp. w/o impact per UG-20(f)	-29°C

Gov. MDMT of the nozzle to shell joint welded assembly : -48°C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18°C



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Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 12.65/51.00 = 0.248

Weld Size Calculations, Description: Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 10.8000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (9.5891 - 17.3325 + 2 \cdot 41.8 \cdot 1.0 \cdot (1.0 \cdot 10.8 - 3.8468)) \cdot 138)$
 $= \max(0, -2716.57) \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_r2) \cdot S_v$
 $= (21.8984 + 1. - 0.0 \cdot 1.0) \cdot 138$
 $= 32221.70 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (21.8984 + 0.0 + 1. + (9.0288)) \cdot 138$
 $= 44926.64 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (21.8984 + 0.0 + 1. + 0.0 + (9.0288)) \cdot 138$
 $= 44926.64 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 273.05 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 29573. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 130.5812) \cdot (45.0 - 3.2) \cdot 0.7 \cdot 138$
 $= 168907. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 273.05 \cdot (6.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 26797. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (29573 + 168907) = 198480 \text{ Kgf}$

Failure Path [Path 2-2]:



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$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (29573 + 0 + 26797 + 0) = 56370 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (\text{Sonw} + \text{Tngw} + \text{Sinw})$$

$$= (29573 + 26797 + 0) = 56370 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 198480 Kgf, must exceed W = 0 Kgf	or W1 = 32221 Kgf
Path 2-2 = 56370 Kgf, must exceed W = 0 Kgf	or W2 = 44926 Kgf
Path 3-3 = 56370 Kgf, must exceed W = 0 Kgf	or W3 = 44926 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 32 bars

The Drop for this Nozzle is : 23.1587 mm.

The Cut Length for this Nozzle is, Drop + Ho + H + T : 287.1587 mm.

Input Echo, WRC297 Item 1, Description: Shell outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.2000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	828.000	mm.
Shell Thickness	14.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.2000	mm.
Nozzle Diameter	332.875	mm.
Nozzle Thickness	45.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 4	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTained loads.

Design Internal Pressure	Dp	12.65	bars
Radial Load	P	-683.25	Kgf
Circumferential Shear	Vc	1019.78	Kgf
Longitudinal Shear	Vl	1019.78	Kgf
Circumferential Moment	Mc	764.54	Kg-m.
Longitudinal Moment	Ml	1019.39	Kg-m.
Torsional Moment	Mt	1274.24	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dn/Tn (7.397) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



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Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 332.875 + 2 \cdot 1.65 \cdot \sqrt{422.6(14.0 - 3.2)}$$

$$= 555.816 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 845.200 mm.
 Nozzle Outside Diameter (d) = 332.875 mm.
 Vessel Thickness used (T) = 10.800 mm.
 Nozzle Thickness used (t) = 41.800 mm.
 T / t = 0.258
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.484$$

Nr/P = 0.054
 Mr/P = 0.092
 M0/P = 0.029
 N0/P = 0.063
 MrD/Mc = 0.260
 NrDL/Mc = 0.092
 M0d/Mc = 0.078
 N0DL/Mc = 0.086
 MrD/Ml = 0.077
 NrDL/Ml = 0.053
 M0D/Ml = 0.022
 N0DL/Ml = 0.070

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-34	34	34	-34
Normal Pressure Stress	48	49	48	49
Outplane Stress Summary	9	59	114	26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-13	-13	13	13
Inplane Bending (ML)	-119	119	119	-119
Inplane Pressure Stress	24	24	24	24
Inplane Stress Summary	-73	100	191	-109

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	6	6	6	6

Shear Stress Summary	8	8	4	4
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	84	102	192	136
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-16	-16	16	16
Outplane Bending (Mc)	-90	90	90	-90
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	24	24	24	24

Outplane Stress Summary	-69	91	144	-56
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	31	-31	31	-31
Inplane Membrane (Mc)	-17	-17	17	17
Inplane Bending (Mc)	-301	301	301	-301
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	48	49	48	49

Inplane Stress Summary	-235	304	402	-262
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	6	6	6	6

Shear Stress Summary	4	4	8	8
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	236	304	402	263
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the	Top	Top	Bottom	Bottom
hoop direction)	Outside	Inside	Outside	Inside



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Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5

Hoop Stress Summary	-10	-9	25	27
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-4	-4	4	4
Axial Bending (ML)	-5	5	5	-5
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-5	1	12	0
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress)

	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1

Shear Stress Summary	2	2	1	1
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	11	11	25	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses in the
 hoop direction)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-16	-16	16	16
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	5	3	5

Hoop Stress Summary	-9	-7	24	25
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	-3	-3	3	3



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Axial Bending (Mc)	-16	16	16	-16
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-15	14	23	-12
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	1	1	1	1
Shear Stress Summary	1	1	2	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	16	21	25	38

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	48	49	48	49	48	49	48	49
Circ. Pl (SUS)	-15	-15	21	21	-14	-14	20	20
Circ. Q (SUS)	-24	24	44	-44	-270	270	332	-332
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	-10	-10	16	16	-13	-13	19	19
Long. Q (SUS)	-88	88	150	-150	-80	80	100	-100
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	48.0	49.0	48.0	49.0	48.0	49.0	48.0	49.0
Pm+Pl (SUS)	33.1	34.0	69.0	70.0	34.0	35.0	68.0	69.0
Pm+Pl+Q (Total)	84.2	103.1	190.3	136.4	236.1	305.1	400.2	263.2

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	49.00	138.00	Passed
Pm+Pl (SUS)	70.03	207.00	Passed
Pm+Pl+Q (TOTAL)	400.19	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):



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Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	5	3	5	3	5	3	5
Circ. Pl (SUS)	-15	-15	21	21	-13	-13	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	-4	-4	4	4	-3	-3	3	3
Long. Q (SUS)	-4	4	6	-6	-15	15	17	-17
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0
Pm+Pl (SUS)	12.0	10.0	24.0	26.0	10.0	8.0	22.0	24.0
Pm+Pl+Q (Total)	12.2	11.2	24.1	27.1	17.1	21.1	22.6	37.1

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	138.00	Passed
Pm+Pl (SUS)	26.00	207.00	Passed
Pm+Pl+Q (TOTAL)	37.05	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input Echo, Tubesheet Number 1, Description: Fixed tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	12.65 bars
Shell Maximum Operating Pressure	Psox,max	12.65 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	14.0000 mm.
Shell Internal Corrosion Allowance	cas	3.2000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	828.000 mm.
Shell Temperature for Internal Pressure	Ts	143.33 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Chnl. rear

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	5.52 bars
Channel Maximum Operating Pressure	Ptox,max	5.52 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	12.0000 mm.
Channel Internal Corrosion Allowance	cac	3.2000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	828.000 mm.
Channel Design Temperature	TEMPC	71.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	370
Tube Wall Thickness	et	2.5200 mm.
Tube Outside Diameter	D	25.4000 mm.
Total Straight Tube Length	Lt	6096.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		5996.00 mm.
Design Temperature of the Tubes		143.33 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	158.80 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm.
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.5200 mm.
Groove Weld Leg	ag	2.5200 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	392.931 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.



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Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	8.0000 mm.
Area of Tube Layout	Ap	3240.3 cm ²
Perimeter of Tube Layout	Cp	2271.6 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Gasketed both Sides, Conf. d

Tubesheet Design Metal Temperature	T	143.33 °C
Tubesheet Material		SA-266 4

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03017
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	100.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	3.2000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	3.2000 mm.
Tubesheet Outside Diameter	A	1020.000 mm.

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	894.00 mm.
Step 1 Depth on the Tubeside	ht1	8.00 mm.
Step 2 Diameter on the Tubeside	dt2	825.00 mm.
Step 2 Depth on the Tubeside	ht2	8.00 mm.
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	894.00 mm.
Step 1 Depth on the Shellside	hs1	8.00 mm.
Step 2 Diameter on the Shellside	ds2	825.00 mm.
Step 2 Depth on the Shellside	hs2	8.00 mm.
Calculated Tubesheet Diameter per Div 2, 4.18.4		894.00 mm.

Note: Tubesheet diameter is now: 894.000 mm. Div 2, 4.18.4.

Dimension G for the Channel Side	Gc	873.151 mm.
Area of the Untubed Lanes	AL	0.0 cm ²

Junction Stress Reduction option	Perform Plastic Calculation
Perform Differential Pressure Design	No
Run Multiple Load Cases	Yes

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side	Both	
Flange Outside Diameter	A	1020.000 mm.
Flange Inside Diameter	B	828.000 mm.
Flange Face Outside Diameter	Fod	897.000 mm.
Flange Face Inside Diameter	Fid	828.000 mm.
Gasket Outside Diameter	Go	886.000 mm.
Gasket Inside Diameter	Gi	860.000 mm.
Small end Hub thk.	g0	16.0000 mm.
Large end Hub thk.	g1	24.0000 mm.
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch	Code Sketch 1a	
Gasket Thickness	tg	4.5000 mm.
Full face Gasket Flange Option	Program Selects	

Bolting Information:

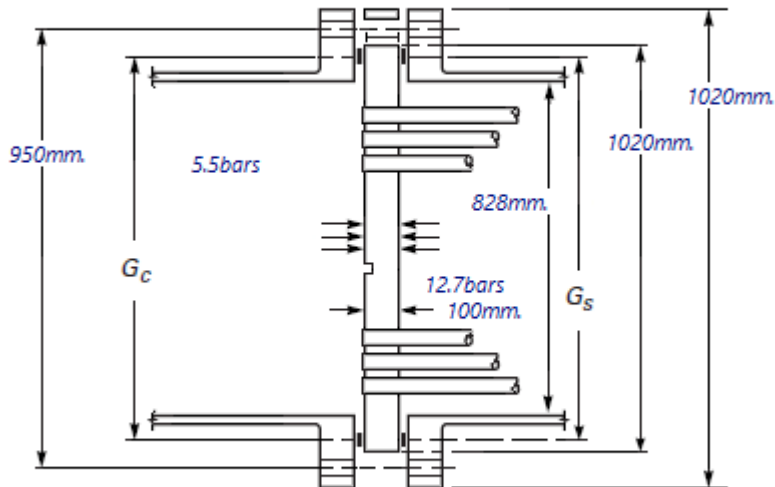
Diameter of Bolt Circle	C	950.000 mm.
Nominal Bolt Diameter	dB	25.4000 mm.
Thread Series	TEMA Thread Series	



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Number of Bolts	n	24
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm.
Alternate Flange Operating Bolt Load, Wm1		90904.12 Kg _f
Alternate Flange Seating Bolt Load, Wm2		123904.77 Kg _f
Alternate Flange Design Bolt Load, W		136768.00 Kg _f
Tubesheet Integral with		None
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	68.0000 mm.
Is Bolt Load Transferred to the Tubesheet		No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	143.3 °C	0.19546E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	71.1 °C	0.19984E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp.	143.3 °C	0.19546E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	143.3 °C	0.19546E+09 KPa.
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Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (5.52 \cdot (25.4/2 - 0)) / (92.4 \cdot 1.0 + 0.4 \cdot 5.52)$
 $= 0.0756 + 0.0000 = 0.0756 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 143.33 °C
 Elastic Modulus for Material 199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.5200	25.40	6388.93	10.08	50.0000	0.0108275	95.84

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 95.8405) / (3 \cdot 10.0794) = 126.7738 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.5210	25.40	6388.93	48.75	50.0000	0.0004628	46.27

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 46.269) / (3 \cdot 48.7531) = 12.6532 \text{ bars}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.5210 mm.
Allowable Internal Pressure at Corroded thickness	199.14 bars
Required Internal Design Pressure	5.52 bars
Allowable External Pressure at Corroded thickness	126.77 bars
Required External Design Pressure	12.65 bars
Required Thickness due to Shell Side pressure	0.5210 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm.
Basic Gasket Width,	$b0 = N / 2$	6.500 mm.
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm.
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	873.156 mm.

Results for ASME U-tube Tubesheet Calculations for Configuration d:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 \cdot ro + dt$
 $= 2 \cdot 392.931 + 25.4$
 $= 811.262 \text{ mm.}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (31.75 - 25.4) / 31.75$
 $= 0.200$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ρ s]:



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= Gs / Do (Configurations d, e, f)
 = 873.1555/811.262
 = 1.0763

Ratio [pc]:
 = Gc / Do (Configurations d)
 = 873.1509/811.262
 = 1.0763

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:
 = ltx / h
 = 0/100.0
 = 0.000 (must be $0 \leq p \leq 1$)

Effective tube hole Diameter [d*]:
 = max(dt - 2tt(EtT/E) • (StT/S) • (p); dt - 2tt)
 = max(25.4 - 2 • 2.52 (0.19546E+09/.19546E+09) •
 (92/137) • (0); 25.4 - 2 • 2.52)
 = 25.4000 mm.

Effective tube pitch [p*]:
 = $p / \sqrt{1 - 4 \cdot \min(A; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) })$
 = $31.75 / \sqrt{1 - 4 \cdot \min(0 ; 4 \cdot 811.262 \cdot 31.75) / (\pi \cdot 811.262^2) })$
 = 31.7500 mm.

Effective ligament efficiency for bending [μ^*]:
 = (p* - d*) / p*
 = (31.75 - 25.4) / 31.75
 = 0.2000

h/p = 3.149606 ; μ^* = 0.200000 ; E* = 30364994 KPa.
 E*/E = 0.155349 ; ν^* = 0.458572

Note: As h/p (3.150) is > 2, data values for h/p = 2 were used.

Skip Step 4 for Configuration d :

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:
 = A / Do
 = 894.0/811.262
 = 1.1020

Coefficient [F]:
 = (1 - ν^*) / E* • (E • ln(K))
 = (1 - 0.46) / 30364994 • (0.19546E+09 • ln(1.1))
 = 0.3385

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:
 = $Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt]$
 = $811.262^2/16 \cdot [(1.076 - 1) \cdot (1.076^2 + 1) \cdot 12.653 -$
 $(1.076 - 1) \cdot (1.076^2 + 1) \cdot 5.516]$
 = 48346.1953 bars•mm.²

Moment acting on unperforated tubesheet rim [M*]
 = Mts + W* • (Gc - Gs) / (2 • π • Do)
 = 48346.2 + 90904.1 • (873.151 - 873.151) / (2 • π • 811.262)
 = 48346.1953 bars•mm.²

Note: W* is the maximum of the bolt loads between the shell and channel sides.



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ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= ((M \cdot) - Do^2/32 \cdot F (Ps - Pt)) / (1 + F)$$

$$= ((48346.2) - 811.262^2/32 \cdot 0.338 (12.65 - 5.52)) / (1 + 0.34)$$

$$= -998.5334 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu) \cdot (Ps - Pt)$$

$$= -998.53 + 811.262^2/64 \cdot (3 + 0.459) \cdot (12.65 - 5.52)$$

$$= 252838.4062 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(| Mp | ; | Mo |)$$

$$= \max(| -998.533 | ; | 252838.406 |)$$

$$= 252838.4062 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 252838.406 / (0.2 (100.0 - 8.0)^2)$$

$$= 89.6218 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138.0$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu \cdot (h - hg')^2)$$

$$= 6 \cdot 252877.797 / (0.2 (60.4431 - 8.0)^2)$$

$$= 275.8550 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc$$

$$= 60.4431 + 0 + 0$$

$$= 60.4431 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB ; HreqS)$$

$$= \max(60.4431 ; 4.6111)$$

$$= 60.4431 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$| Ps - Pt |$]:

$$= | Ps - Pt |$$

$$= | 12.65 - 5.52 |$$

$$= 7.137 \text{ bars}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / Do \cdot \min(0.8 \cdot S ; 0.533 \cdot Sy)$$

$$= 4 \cdot 0.2 \cdot 100.0 / 811.26 \cdot \min(0.8 \cdot 137.997 ; 0.533 \cdot 220.065)$$

$$= 10.887 \text{ N./mm}^2$$

$| Ps - Pt |$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S ; 0.533 \cdot Sy)) \cdot (4 \cdot Ap / Cp) \cdot | Ps - Pt | + Cats + Catc$$

$$= 1 / (4 \cdot 0.2 \cdot 110.398) \cdot (4 \cdot 3240.3 / 2271.59) \cdot | 12.65 - 5.52 | + 0$$

$$= 4.6111 \text{ mm.}$$



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Note: Analysis Completed for Tubesheet Configuration d.

Tube Metal Area [At]:
 $= \pi (d_o - t) t$
 $= \pi (25.4 - 2.52) 2.52$
 $= 1.811 \text{ cm}^2$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:
 $= \pi \cdot t (d_o - t) S$
 $= \pi \cdot 2.52 (25.4 - 2.52) 92.398$
 $= 1706.640 \text{ Kgf}$

Groove Weld Strength [Fg]:
 $= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$
 $= 0.85 \cdot \pi \cdot 2.52 (25.4 + 0.67 \cdot 2.52) 92.4$
 $= 1706.6400 \text{ Kgf}$

Weld Strength Factor [fw]:
 $= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$
 $= 92.398 / 92.398$
 $= 1.000$

Design Strength Ratio [fd]:
 $= 1 \text{ for full strength welds}$

Fillet Weld Strength [Ff]:
 $= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$
 $= 0.55 \cdot \pi \cdot 2.52 (25.4 + 0.67 \cdot 2.52) 92.4$
 $= 1111.3018 \text{ Kgf}$

Minimum Length of Weld [ar]:
 $= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$
 $= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.52 \cdot (25.4 - 2.52) 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$
 $= 3.111 \text{ mm.}$

Minimum Required Fillet Weld Leg [afr]: 1.5557 mm.
 Minimum Required Groove Weld Leg [agr]: 1.5557 mm.

Tube-Tubesheet Jt allowable, 1706.64 is $\geq$ tube strength 1706.64 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	89.6 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	5.1 <=	110.4 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	60.443	100.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.556	2.520	Ok
Tube-Tubesheet Groove Weld Leg :	1.556	2.520	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:



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Case#	--Reqd. Thk. + CA	Bend	Tubesheet		Stresses		Case Type	Pass/Fail
	Tbsht Extnsn		Allwd	Shear	Allwd			
D1uc	54.119	1.074	69	276	4	110	Ps+Pt D1	Ok
D2uc	77.807	2.464	159	276	9	110	Ps+Pt D2	Ok
D3uc	60.443	1.390	90	276	5	110	Ps+Pt D3	Ok
D1c	57.299	1.074	74	276	4	110	Ps+Pt-c D1	Ok
D2c	81.076	2.464	171	276	10	110	Ps+Pt-c D2	Ok
D3c	63.634	1.390	96	276	5	110	Ps+Pt-c D3	Ok
Max:	81.0758	2.464 mm.		0.618		0.087	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Note:

Because there was no net pressure differential across the tubesheet due to the specified pressures psd,min and ptd,min; load case D4 was not processed as the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	81.076	100.000	Ok
Tubesheet thickness flanged extension :	2.464	68.000	Ok
Tube thickness :	0.521	2.520	Ok
Tube-tubesheet fillet weld leg :	1.556	2.520	Ok
Tube-tubesheet groove weld leg :	1.556	2.520	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	20.476	1.000	20.476	1.000
Tubesheet Shear Stress	144.870	1.000	144.870	1.000
Tube Pressure Stress	199.135	1.000	126.773	1.000
Minimum MAWP	20.476		20.476	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	21.978	1.000	21.978	1.000
Tubesheet Shear Stress	154.776	1.000	154.776	1.000



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Tube Pressure Stress	199.135	1.000	126.773	1.000
Minimum MAPnc	21.978		21.978	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Determine the governing MDMT considering the governing condition:

Governing thickness on the shell side per figure UCS-66.3 (c):
= tubesheet thickness/4
= 100.0/4
= 25.000 mm.

Thickness Ratio = 0.576, Temperature Reduction per Fig. UCS 66.1 = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-25 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
= max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
= max(5.52/21.98, 12.65/21.98)
= 0.576

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Channelside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: